GUBELLINI DataAnalysis

User Manual

Telemetry data analysis software · Revision 025 · November 2026



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1 Introduction

GUBELLINI DataAnalysis is the free telemetry data analysis software supplied with **Gubellini Electronics** data loggers. It opens the logger recordings, displays the channels on synchronized graphs, calculates and compares lap times, shows the vehicle position on the circuit map, creates calculated channels, analyzes statistics and frequencies, synchronizes an onboard video and generates reports.

The software is designed to work **even when the GPS signal is missing or intermittent**: distance, laps and map position are reconstructed from alternative channels (fallback speed, lap beacon, logger lap index).

1.1 How this manual is organized

The chapters follow the normal workflow: installation, opening the data, reading the graphs, laps and map, comparison, analysis tools, advanced functions and settings. All the examples and images were made with the **sample dataset** included in the package: a test session at the Medicina (BO) circuit on 14/10/2026.

Sample dataset	Value
File	`Data\LOGGER__20261014_1838_Test_GABMW_000011.dat`
Duration / samples / rate	234.6 s · 2346 samples · 10 Hz
Laps	Out Lap, 4 complete laps, In Lap – best lap: **Lap 4, 00:35.151**
Reference lap length	398 m (Medicina track)
Main channels	GPS_Speed (GPS3), RPM_OBDII (CAN2), TPS_OBDII (CAN3), Acc_X/Acc_Y/Acc_Z (IMU1–3), WATER_TEMP_OBDII (CAN5)…

1.2 Conventions

- **Bold**: names of menus, buttons and windows as they appear in the program (e.g. **File → Open Dataset**).
- `Ctrl`+`O`: keyboard shortcut. The shortcuts shown are the defaults and can be changed (chapter 19).

> **Language**
> The program is available in Italian, English, French, Spanish, German and Portuguese (**Edit → Preferences → Language**). The images in this manual were captured with the English interface and the dark theme.

2 Installation and startup

2.1 Requirements

- Windows 10 or 11, 64-bit, 8 GB of RAM recommended, screen of at least 1600×900.
- Internet connection only for checking for updates and for the AI assistant with cloud providers (both optional).

2.2 Installing with the installer

1. Run `GUBELLINI_DataAnalysis_Setup_Rev025.exe` and follow the wizard.
2. The default folder is `C:\Program Files\GUBELLINIDataAnalysis`; you can choose to create a desktop icon.
3. When the installation is complete, the program can be started directly from the last page of the installer or from the Start menu.

Besides the program, the installation folder contains:

Folder	Contents
`Data`	Medicina sample dataset with its companion folder, plus five more sample sessions with GPS and laps: EasyKart60 (Ottobiano), Jaguar (Misano), Porsche 997 (Daytona), SuperKart (Nürburgring), Yamaha R6 (Imola).
`TrackMap`	Maps of 5265 circuits in CSV format, georeferenced, with the `track_maps.xml` catalog.
`Layouts`	Channel layouts, preferences (`app_state.json`), shortcuts, math channels.
`LookupTables`	2D lookup tables.
`Branding`	Logo used in the reports.
`Reports` , `ADVCompareLayouts`	Generated reports and saved ADV Compare layouts.

> **Updating without losing your settings**
> When a newer revision is installed, the installer **does not overwrite** the `Layouts` folder: the user's layouts, preferences, shortcuts and math channels are preserved. In Windows the program always appears as **GUBELLINI DataAnalysis**, without the revision number: every update replaces the previous installation instead of adding a new application. The installed revision is shown in the title of the main window and in **Help → About GUBELLINI DataAnalysis**.

2.3 Portable version

As an alternative to the installer, extract the `.zip` package into a folder and double-click `run.bat`. The script checks that Python and the required libraries are present and starts the program.

2.4 First startup

At first startup the window is empty: the central graph contains no data and the side windows are ready to be filled when a dataset is opened.

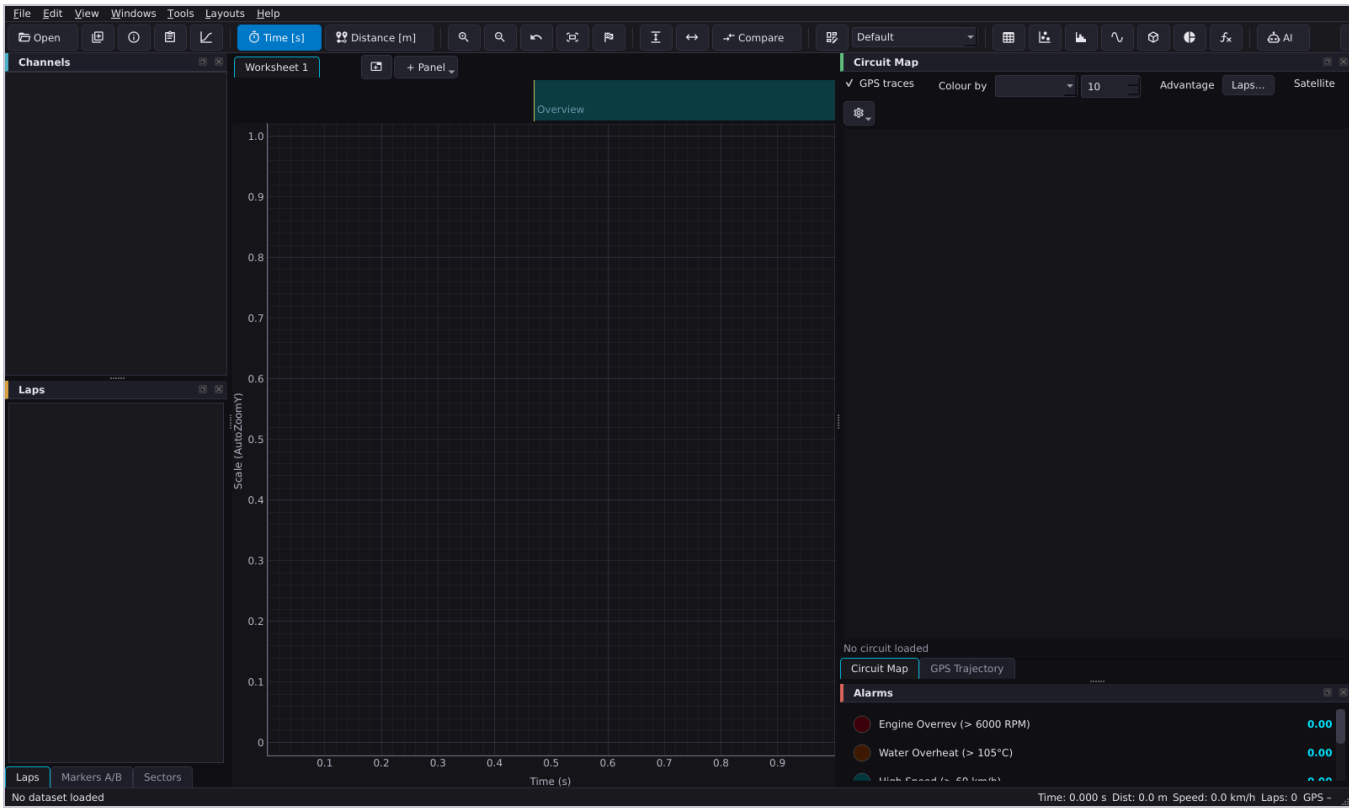


Figure 1 – The main window at first startup, with no data.

3 The interface

The main window consists of a central area with the graphs and of dockable windows (*docks*) that can be moved, placed side by side, grouped into tabs or detached from the window. Their size and arrangement are remembered when the program is closed.

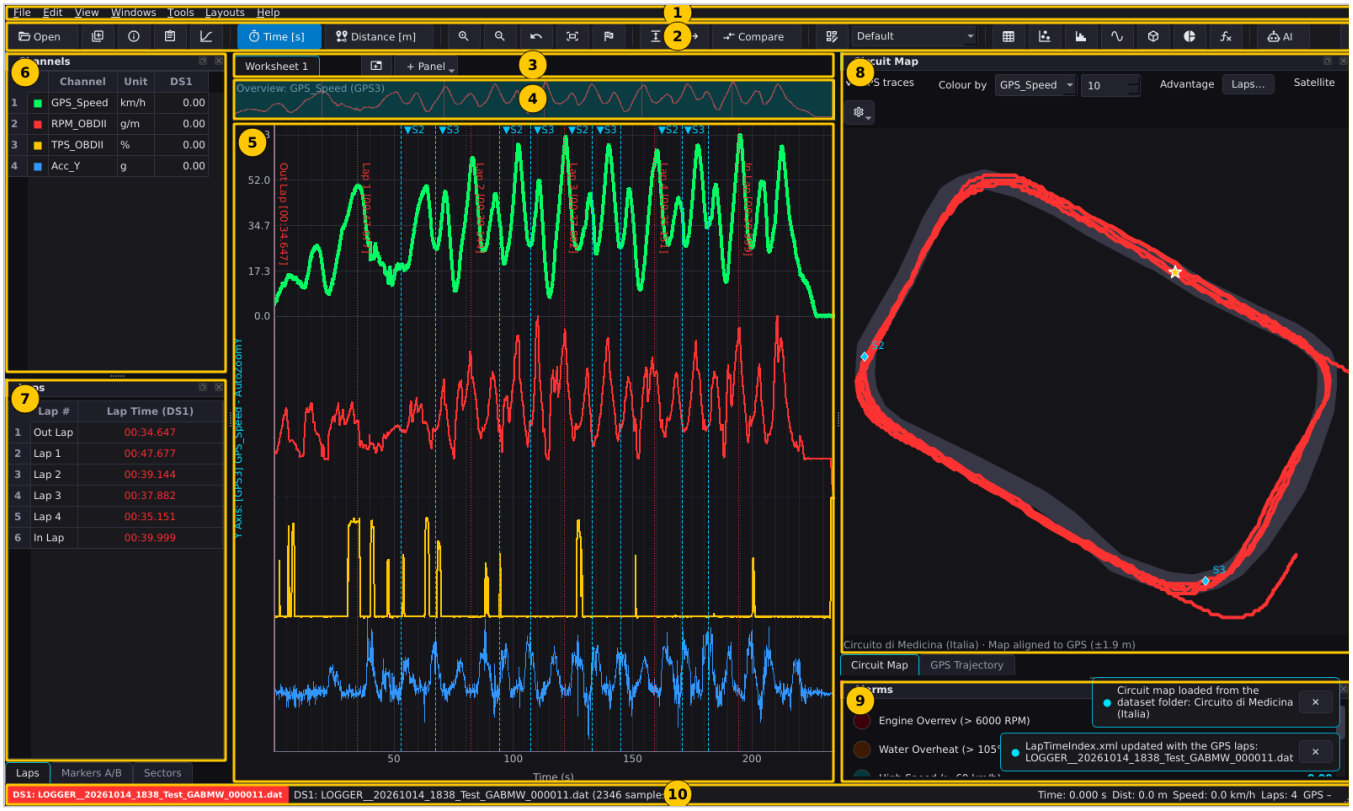


Figure 2 – The areas of the main window with the sample dataset loaded.

1. **Menu bar**: File, Edit, View, Windows, Tools, Layouts, Help.
2. **Toolbar**: open dataset, time/distance base, zoom, AutoZoomY, shift, comparison, layout and analysis windows.
3. **Worksheets**: work tabs, each with its own graphs (chap. 10).
4. **Overview**: the whole session in miniature, with a frame showing the displayed part.
5. **Main graph**: the channels of the active layout, stacked in bands (*slots*).
6. **Channel Values Table**: value of each channel under the cursor, for each dataset.
7. **Lap Times and Outings**: list of laps; the **Markers A/B** tab is in the same area.
8. **Circuit Map** and **GPS Trajectory Map**.
9. **Virtual LEDs / Alarms**.
10. **Status bar**: buttons of the loaded datasets and time, distance, speed, laps and GPS status at the cursor.

3.1 Status bar

On the left there is a colored button for each dataset: clicking it makes that dataset the **selected** one (used by shift, close, statistics…). On the right, for the cursor position, the bar shows **Time**, **Dist**, **Speed**, the number of laps and **GPS ✓** / **GPS ✗**, which indicates whether the GPS fix is valid at that point.

3.2 Notifications

Non-critical information (map loaded from the dataset folder, lap index updated, dataset added…) appears as notifications at the bottom right, which close by themselves or with ×, without interrupting your work.

3.3 Dockable windows

- Drag the title bar of a window to move it; dropping it onto another window creates a tabbed group.
- The button to the left of × detaches the window into an independent frame (useful with two monitors).
- Closed windows can be reopened from the **Windows** menu.
- **Layouts → Save Windows Layout** stores the arrangement; **Restore Default Windows Layout** returns to the initial one.
- The windows on the right side (map, LEDs, AI assistant…) can be made very narrow: their buttons wrap onto more rows.

3.4 Light and dark theme

The dark theme is the default. `Ctrl`+`Shift`+`L` (or **View → Toggle Dark / Light Theme**) switches to the light theme, better suited to printing or to very bright environments.

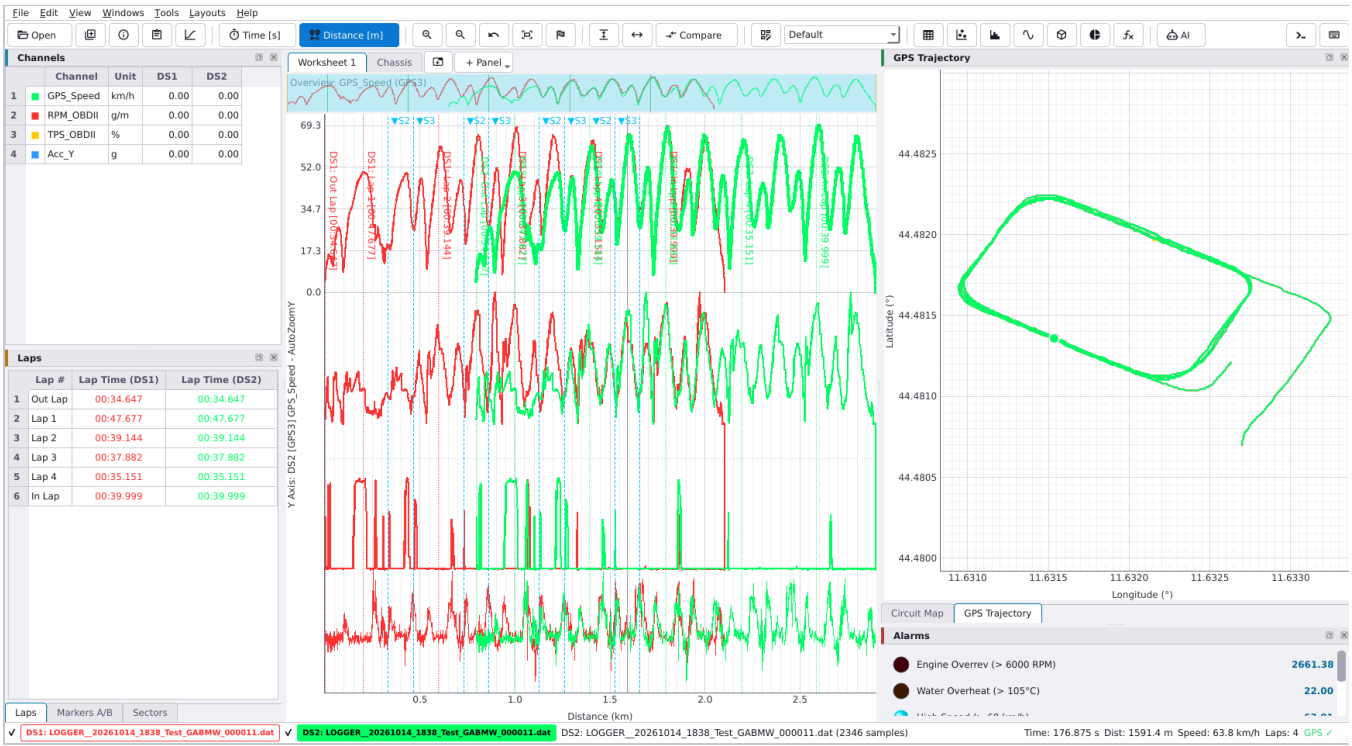


Figure 3 – The light theme, with two datasets and two worksheets.

4 Opening data

4.1 Opening a dataset

1. **File → Open Dataset (.dat)…** (`Ctrl`+`O`) or the **Open Dataset** button.
2. In the **Open dataset** window choose the `.dat` file (see below). The window starts from the last folder used.
3. The program reads the channels, filters the GPS, calculates distance and laps and applies the active layout.

Loading window. After the dataset is confirmed, a window lists the loading steps – reading the file, GPS filter and coherence check, distance calculation, lap detection, session distance correction, math channels, drawing graphs and tables, circuit map – with the step in progress, the duration of the completed ones and the elapsed time. The calculations run in the background, so the window stays updated also with large files; it closes by itself when the dataset is ready. It also appears when a change in the preferences requires reloading the datasets.

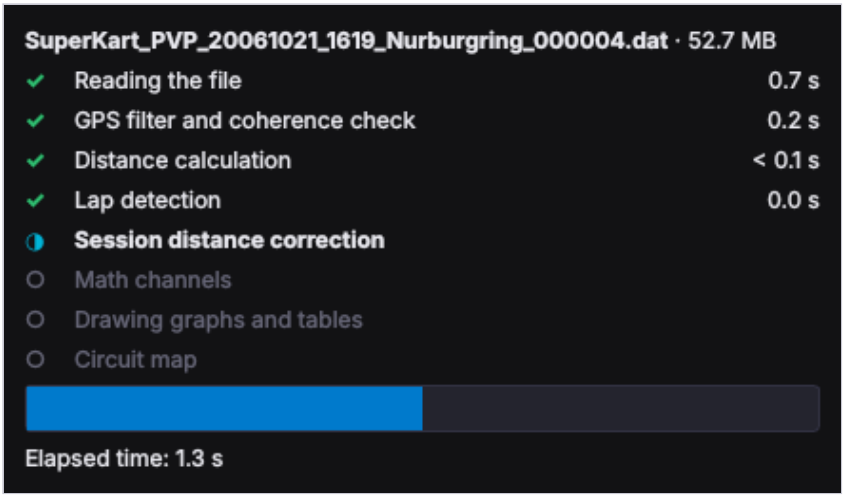


Figure 4 – The loading window during the opening of a 53 MB dataset.

Open window. On the left you browse the folders of the PC (**Up** button, editable path, list with **Last folder used**, **Sample data**, **Home folder** and the drives, **Browse…** button). Only the `.dat` files that have their companion folder (with circuit map, LapTimeIndex.xml, datasheet…) are listed; the companion folders are not shown. Untick **Only datasets with their folder** to see every `.dat` file.

Clicking a dataset shows on the right the **circuit** (when the folder has the map), the **session date**, the logger with the dataset number, duration and number of samples, and the **datasheet** (when present). By default all the groups of the datasheet are shown; **Choose entries** shows the whole tree with a checkbox for each entry: tick the ones to show and press **Choose entries** again. The choice is remembered and used for every dataset. Double-click the dataset or press **Open** to open it. The preview reads only the header of the file, so it appears immediately also for large files.

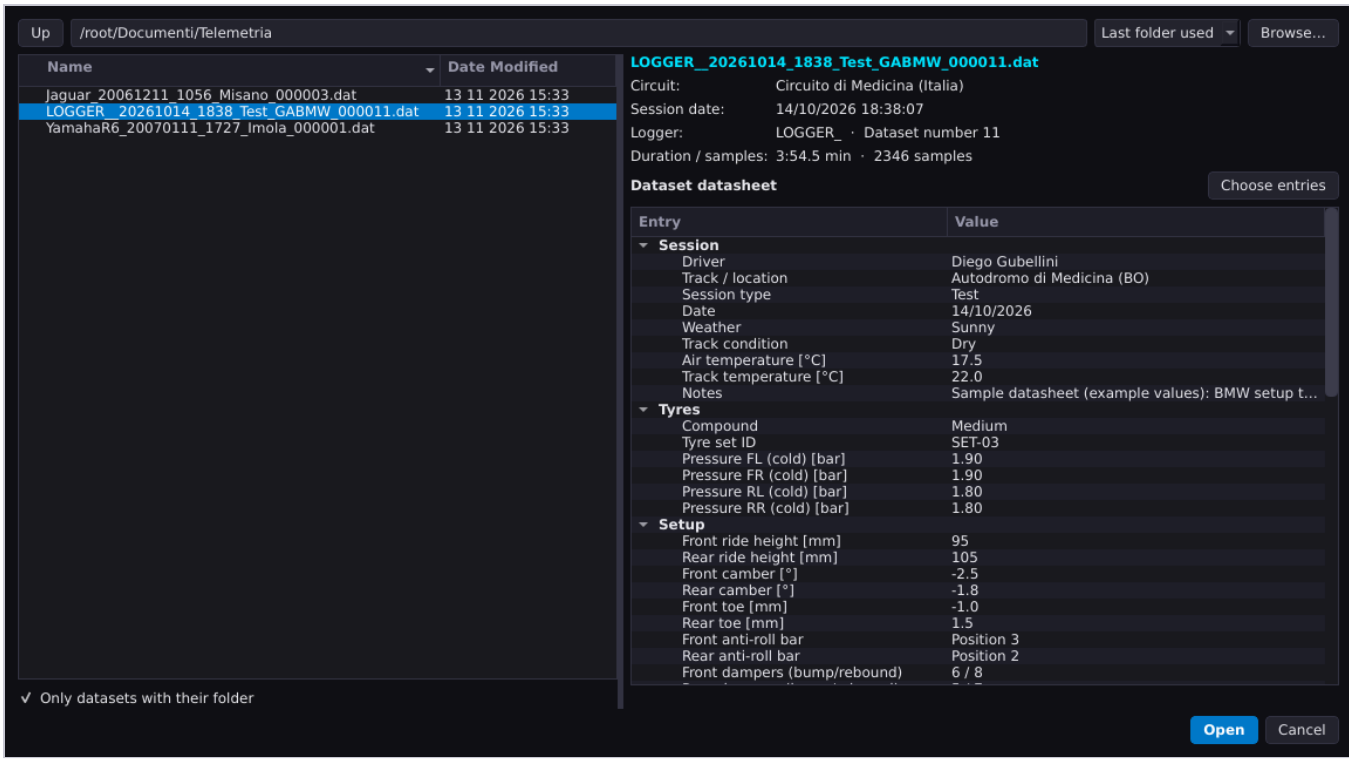


Figure 5 – The Open dataset window: folders with the datasets only on the left, circuit, date and datasheet of the selected dataset on the right.

Alternatively, you can **drag** the file onto the window or reopen it from **File → Recent Files**.

Large files. Sessions of 100 000 samples and more (e.g. 20 minutes at 100 Hz) stay fluid: the graphs draw only the visible part, reduced to the screen resolution while keeping peaks and valleys, so moving the cursor, zooming and scrolling are immediate.



Figure 6 – The Medicina dataset just opened: graphs, laps, circuit map aligned to the GPS and alarms.

4.2 The dataset companion folder

Next to the `.dat` file, the logger saves a folder with the same name containing additional information. GUBELLINI DataAnalysis uses it automatically:

File	Use
`Channels_details.xml`	Name of each channel (e.g. GPS3 → **GPS_Speed**, CAN2 → **RPM_OBDII**). These labels are used throughout the program: values table, axes, statistics, export, reports and AI assistant.
`Channels_setting.xml`	Configuration of the logger channels (rates, factors).
`LapTimeIndex.xml`	Lap index: sample number of each finish line crossing (chap. 6).
`*_track.csv`	Circuit map: it is loaded and aligned to the GPS automatically (chap. 7).

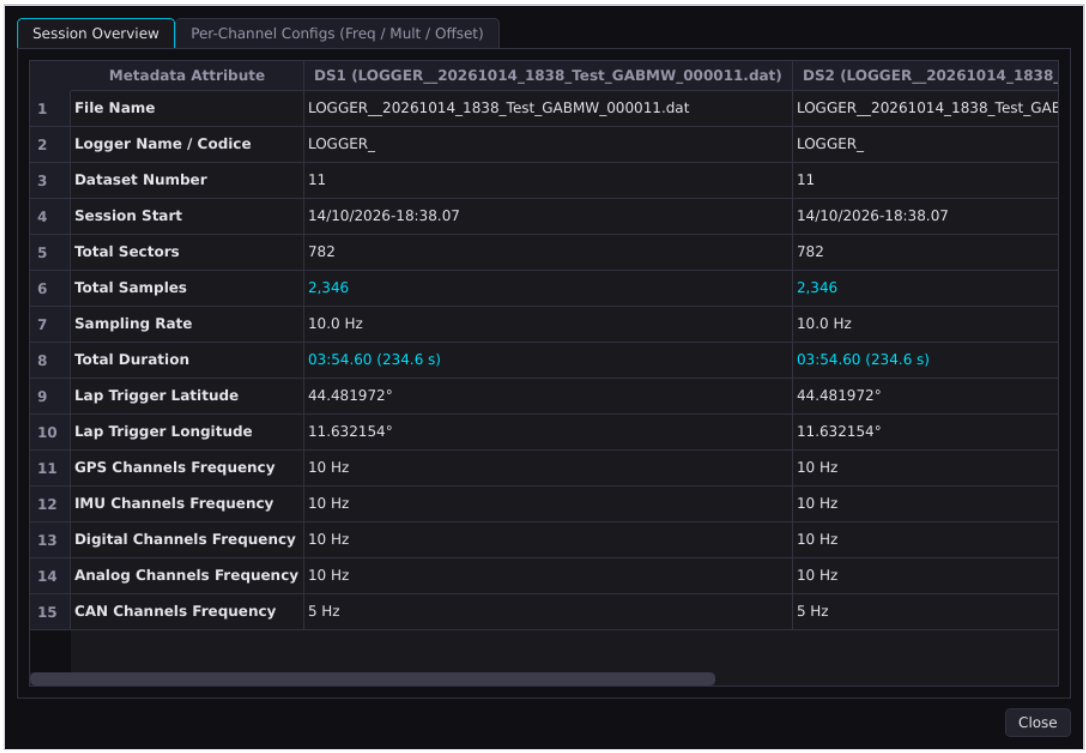
> **Example**
> When the sample dataset is opened, two notifications appear: *«Circuit map loaded from the dataset folder: Medicina_circuit_track.csv»* and *«LapTimeIndex.xml updated with the GPS laps»*. The channel names in the table (GPS_Speed, RPM_OBDII, TPS_OBDII, Acc_Y) come from `Channels_details.xml`.

4.3 Multiple datasets

File → Add Dataset… (`Ctrl`+`Shift`+`O`) loads another file without closing the open ones (up to 5). Each dataset has a color (DS1 red, DS2 green, DS3 blue…, which can be changed in the Preferences) and is drawn overlaid on the others in all the graphs. **File → Close Selected Dataset** (`Ctrl`+`W`) closes the dataset selected in the status bar. Keys `1`…`5` select the corresponding dataset. **Dataset numbers:** every dataset keeps its number (and its color) until it is closed; a new dataset, opened with **Add Dataset**, takes the first free number. For example, with DS1 and DS2 open, closing DS1 and adding a new file makes it DS1 while DS2 stays open; with DS1 and DS3 open the new file becomes DS2. Any dataset can be closed, DS1 included. **Open Dataset** replaces the selected dataset, which keeps its number (with no dataset open it becomes DS1); only **Add Dataset** adds one (§24.2).

4.4 Dataset information

File → Dataset Information… (`Ctrl`+`I`) shows the metadata of each open dataset: logger name, start date and time, number of samples, rate, duration, finish line coordinates and rates of the channel groups. The **Per-Channel Configs** tab lists the rate, multiplier and offset of each channel.



	Metadata Attribute	DS1 (LOGGER__20261014_1838_Test_GABMW_000011.dat)	DS2 (LOGGER__20261014_1838_
1	File Name	LOGGER__20261014_1838_Test_GABMW_000011.dat	LOGGER__20261014_1838_Test_GAE
2	Logger Name / Codice	LOGGER_	LOGGER_
3	Dataset Number	11	11
4	Session Start	14/10/2026-18:38.07	14/10/2026-18:38.07
5	Total Sectors	782	782
6	Total Samples	2,346	2,346
7	Sampling Rate	10.0 Hz	10.0 Hz
8	Total Duration	03:54.60 (234.6 s)	03:54.60 (234.6 s)
9	Lap Trigger Latitude	44.481972°	44.481972°
10	Lap Trigger Longitude	11.632154°	11.632154°
11	GPS Channels Frequency	10 Hz	10 Hz
12	IMU Channels Frequency	10 Hz	10 Hz
13	Digital Channels Frequency	10 Hz	10 Hz
14	Analog Channels Frequency	10 Hz	10 Hz
15	CAN Channels Frequency	5 Hz	5 Hz

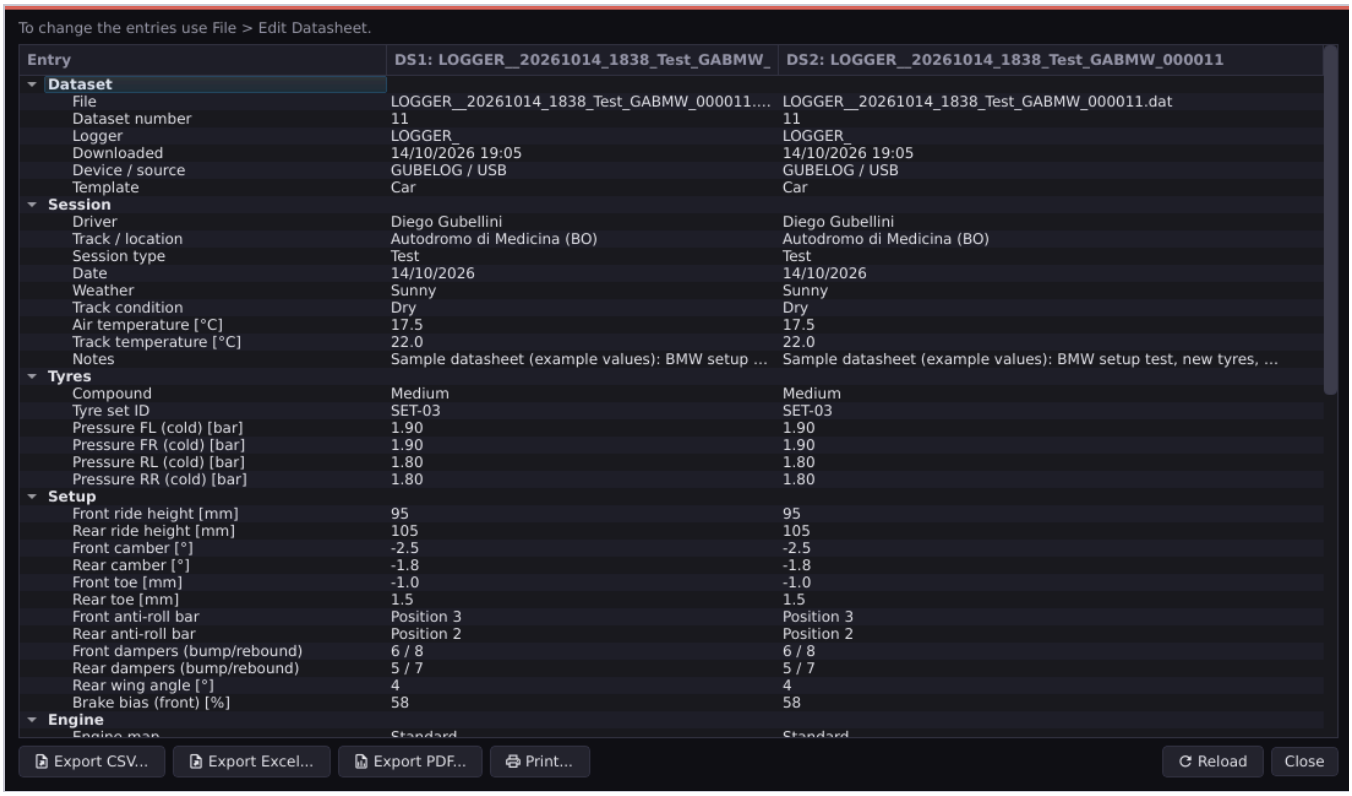
Figure 7 – Dataset information with two datasets open.

4.5 Dataset datasheet

At the end of every download **GUBELLINI DataStudio** saves a `Datasheet.xml` file in the dataset folder: the entries of the sheet (driver, circuit, setup…) organized in groups, and the **special entries**, i.e. the parts monitored by distance or by hours of use (engine, chain…) with their counters updated by the download.

File → Dataset Datasheet... (`Ctrl`+`Shift`+`D`, or the icon next to Dataset Information on the toolbar) opens a table with one column for each open dataset that has a datasheet:

- **Dataset**: file, dataset number, logger, download date, device / source and template used;
- the **groups of entries** of the template with the values entered at the download;
- **Special entries (parts)**: channel the usage is computed from, distance and time of use of the dataset and, for each part, the contribution of this dataset, the total, the threshold and the status – **OK**, **Near the threshold** or **Threshold exceeded** – with the percentage of the threshold reached.



Entry	DS1: LOGGER__20261014_1838_Test_GABMW_	DS2: LOGGER__20261014_1838_Test_GABMW_000011
Dataset		
File	LOGGER__20261014_1838_Test_GABMW_000011....	LOGGER__20261014_1838_Test_GABMW_000011.dat
Dataset number	11	11
Logger	LOGGER_	LOGGER_
Downloaded	14/10/2026 19:05	14/10/2026 19:05
Device / source	GUBELOG / USB	GUBELOG / USB
Template	Car	Car
Session		
Driver	Diego Gubellini	Diego Gubellini
Track / location	Autodromo di Medicina (BO)	Autodromo di Medicina (BO)
Session type	Test	Test
Date	14/10/2026	14/10/2026
Weather	Sunny	Sunny
Track condition	Dry	Dry
Air temperature [°C]	17.5	17.5
Track temperature [°C]	22.0	22.0
Notes	Sample datasheet (example values): BMW setup ...	Sample datasheet (example values): BMW setup test, new tyres, ...
Tyres		
Compound	Medium	Medium
Tyre set ID	SET-03	SET-03
Pressure FL (cold) [bar]	1.90	1.90
Pressure FR (cold) [bar]	1.90	1.90
Pressure RL (cold) [bar]	1.80	1.80
Pressure RR (cold) [bar]	1.80	1.80
Setup		
Front ride height [mm]	95	95
Rear ride height [mm]	105	105
Front camber [°]	-2.5	-2.5
Rear camber [°]	-1.8	-1.8
Front toe [mm]	-1.0	-1.0
Rear toe [mm]	1.5	1.5
Front anti-roll bar	Position 3	Position 3
Rear anti-roll bar	Position 2	Position 2
Front dampers (bump/rebound)	6 / 8	6 / 8
Rear dampers (bump/rebound)	5 / 7	5 / 7
Rear wing angle [°]	4	4
Brake bias (front) [%]	58	58
Engine		
Engine map	Standard	Standard

Figure 8 – Datasheet of two open datasets: the gearbox has exceeded its 40-hour threshold, the engine oil and the front brake pads are near their thresholds.

The values of the entries can be changed with **File → Edit Datasheet** (§25.1): the file keeps the DataStudio format and its part counters. To reset a counter or change the template use GUBELLINI DataStudio; **Reload** reads the files again.

The table can be exported with **Export CSV...** (`;` separator, opens in Excel), **Export Excel...** (`.xlsx` file with coloured statuses) and **Export PDF...**, or printed with **Print...**. With more than two datasets the PDF and the printout are in landscape.

> **Example**
>
> The Medicina sample dataset has a datasheet created with the *Car* template of GUBELLINI DataStudio (example values): the Session, Tyres, Setup and Engine groups and six parts – Engine (31.3 h of 40 h, OK), Engine oil (1852 km of 2000 km, near the threshold), Gearbox (40.4 h of 40 h, threshold exceeded), Clutch, Front brake pads (near the threshold) and Front brake discs. The dataset added 2.1 km and 0.065 h of use.

5 The graphs

5.1 Cursor and reading values

When the mouse moves over the graph, the vertical cursor follows the pointer; a click fixes it. The cursor is synchronized with all the graphs, with the map (red dot), with the channel values table and with the video. From the keyboard:

Keys	Action
← →	Cursor by one sample (with Shift by 10 samples)
Home / End	Start / end of the data
PgUp / PgDown	Pans one screen left / right
Ctrl+← / Ctrl+→	Previous / next lap

"Graph" shortcuts work when the mouse is over a graph: there is no need to click.

5.2 Time base and distance base

The horizontal axis can be **time** in seconds (T, **Time Base [s]** button) or the **distance** traveled (D, **Distance Base [m]**). In distance base, laps from different datasets overlap corner by corner even if their times differ: this is the recommended base for comparisons.



Figure 9 – Lap 4 in time base.



Figure 10 – The same lap in distance base.

> **How distance is calculated**
>
> Distance is the integral of the strategic speed (normally GPS_Speed). When the GPS is not valid, the **fallback speed channel** (e.g. wheel speed from CAN, set in the Preferences) or an interpolation is used. With **distance correction** enabled, each complete lap is rescaled to the reference lap length and, when the GPS is valid, the position within the lap is projected onto the reference trajectory: this way the odometer does not accumulate drift from one lap to the next.

5.3 Zoom

Command	Action
Mouse wheel	Zoom in/out centered on the pointer
+ / -	Zoom in / out
Backspace	Undoes the last zoom
F	Shows the whole dataset
L	Zooms on the lap where the cursor is
Dragging with the left button	Pans the graph
Dragging with the **right** button	Selects an interval: on release, **Zoom Selected Interval** appears
Frame in the overview	Drag it to move the displayed window; its edges change the width



Figure 11 – Selecting an interval with the right button: the highlighted area becomes the new view.

A right-click without dragging opens the standard graph menu (export, axis settings).

5.4 Vertical scale: AutoZoomY

With **AutoZoomY** (Y) the vertical scale of each channel adapts to the values visible in the window; in **ManualZoomY** the Min/Max limits set in the layout are used (chap. 8). Clicking a channel in the **Channel Values Table** highlights it (thicker line) and the Y axis on the left shows its scale and label. The mode applies to all the graphs of the worksheet: main graph and additional graphs (also when it is changed from the context menu of a graph).

5.5 Overview

The strip above the graph shows the whole session of the strategic speed channel (e.g. GPS_Speed) or of the last channel clicked in the table. The light frame indicates the part visible in the main graph and is always synchronized. This also applies to the channels of the additional graphs: every channel clicked in the table is shown in the overview. When laps are detected, the start of every lap is marked in the overview too, with a vertical line in the dataset color at 50% transparency.

5.6 Markers A/B and measurements

Place the cursor and press A, then move and press B: two draggable dashed lines appear. The **Markers A/B** tab shows Δt, Δd, average speed and, for each channel, the value at A and at B and the minimum, maximum and average between the two markers. C clears the markers.



Figure 12 – Markers A and B on a section of the lap and the Markers A/B tab with the measurements.

> **Example – duration of a braking phase**
>
> Place A at the start of the drop in GPS_Speed before a corner and B at the point of minimum speed: the Markers A/B tab shows the braking time (Δt), the braking distance (Δd) and the minimum speed reached.

5.7 Channel Values Table

Lists the channels of the layout with color, name, unit of measurement and the value at the cursor for each dataset (DS1, DS2…). The value takes the dataset shift into account.

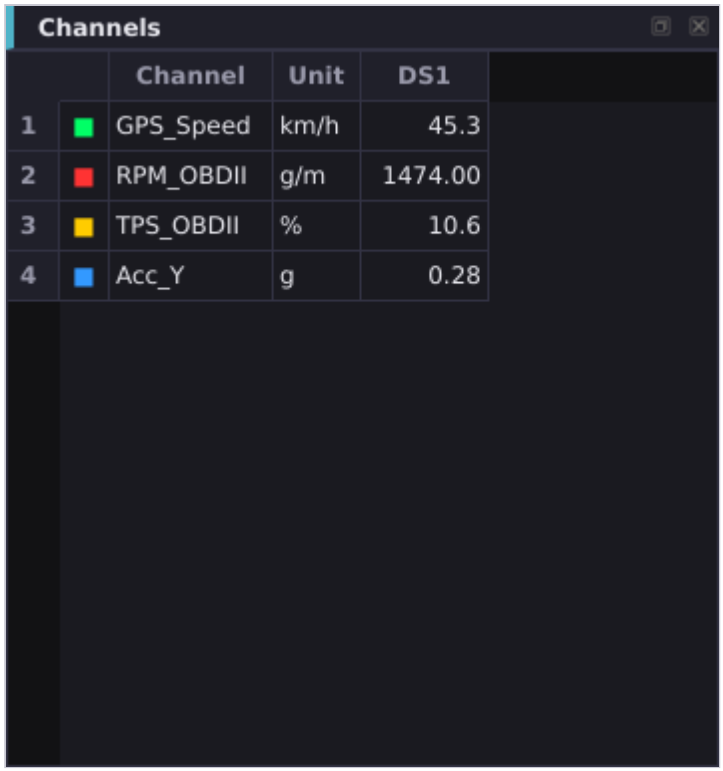


Figure 13 – The Channel Values Table.

Show/Hide Min-Max. Right-click a channel and choose **Show Min-Max**: the relative minima and maxima of the channel in the visible part of the graph are marked with a small arrow and the value (with the decimals of the layout). A point is a relative maximum or minimum when the signal moves back by at least 15% of its visible range before the next opposite extreme: on a speed trace these are the ends of the straights and the apexes of the corners. With two or more datasets the extremes of every dataset are shown in its color and the texts are moved so that they do not overlap. The marks follow zoom and scrolling; **Hide Min-Max** removes them. The text size is set in **Preferences → Font sizes**.

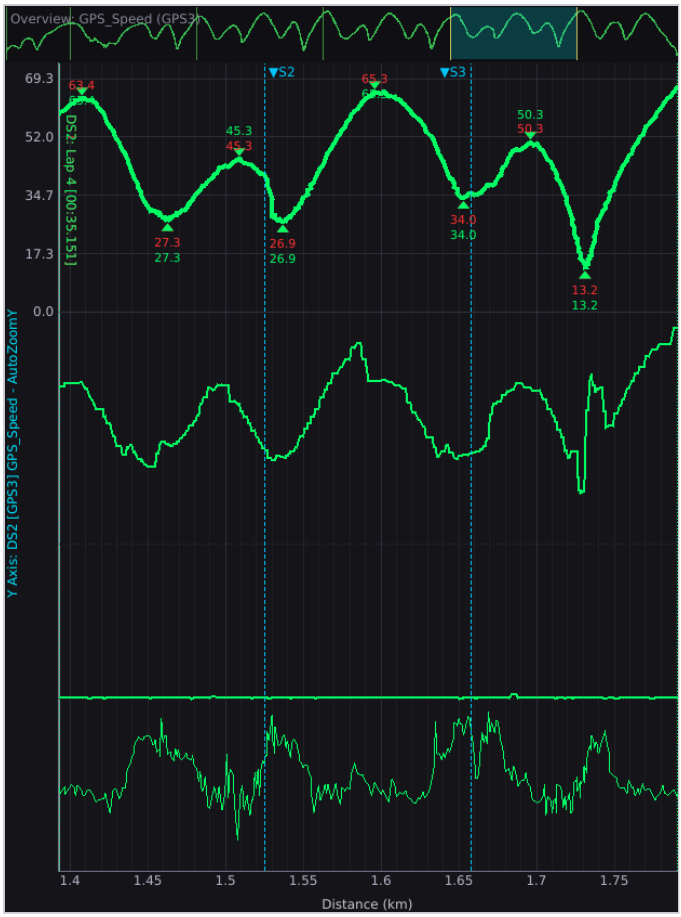


Figure 14 – Min-Max of GPS_Speed on the best lap with two datasets.

5.8 Screenshots and printing

File → Save Graph Screenshot… (`Ctrl`+`Shift`+`S`) saves an image of the graphs; **File → Print Graph…** (`Ctrl`+`P`) sends it to the printer.

6 Laps and lap times

6.1 Lap Times and Outings window

Lists the laps of each dataset: **Out Lap** (leaving the pits), numbered complete laps and **In Lap** (returning to the pits). The best lap of each dataset is highlighted. Only complete laps are used for the best lap, comparisons and the reference lap.



Figure 15 – Lap times of the sample dataset.



Figure 16 – Lap Time Details: distance, maximum and average speed.

- **Double-click** on a time: with a single dataset the view moves to the start of the lap; with several datasets only the dataset of that lap is shifted, so that the lap starts at the left edge of the view (useful for overlaying laps from different datasets).
- **Right-click** on a time: **Fit zoom to this lap** (zoom on the lap) and **Move at beginning of this lap** (cursor at the start of the lap).
- **Lap Time Details…**: complete table with distance, maximum and average speed of each lap.

6.2 How laps are detected

The lap source is chosen in **Preferences → Lap Trigger Source**:

Source	How it works
Extract from Dataset Metadata	GPS finish line with the coordinates saved by the logger (default).
Manual Geographic Coordinates	GPS finish line with the latitude and longitude entered in the Preferences and a detection radius.
Lap beacon / digital channel (no GPS)	A rising edge of a channel (e.g. beacon receiver) above the threshold marks the crossing. Works without GPS.
Logger lap index (LapTimeIndex.xml)	Uses the crossings written in the `LapTimeIndex.xml` file of the dataset folder.

The crossing instant is interpolated between two samples for greater accuracy. The **Minimum lap time** prevents double crossings.

> **No GPS**
> If the dataset has no valid GPS fix and the source is GPS, when it is opened you are offered to **import** a LapTimeIndex.xml file or to create one with the **wizard** (§6.4). Alternatively, a beacon channel can be used.

6.3 LapTimeIndex.xml

This is the logger file with the lap index: the first value is 0 (start of recording), the following ones are the sample numbers of the finish line crossings; unused slots are set to −1.

- When the laps are calculated with the **GPS trigger**, the file is **updated** with the GPS crossings. At the first change, the logger's original is kept as `LapTimeIndex.xml.bak`.
- **Tools → Import LapTimeIndex.xml** loads an index from another file.

6.4 Lap index wizard

It is used to create the lap index when the GPS is missing. It is opened from **Tools → Lap index wizard (LapTimeIndex.xml)**, from the command palette or from the prompt offered when the dataset is opened.

1. **1. Show the whole dataset**: the graph switches to distance base and shows the whole session.
2. **2. Offset**: distance of the first finish line crossing. You can type it, drag the orange *Offset* line on the graph or press **Use cursor**.
3. **3. Lap length**: length of the lap. **From circuit map** takes it from the georeferenced circuit map. A dashed line *Lap 1, Lap 2...* appears on the graph at each lap, and the table lists the sample, time and start of each lap.
4. **4. Confirm and generate XML**: the `LapTimeIndex.xml` file is written to the dataset folder, in the same format as the logger's, and the laps are recalculated.



Figure 17 – The wizard: offset 195 m, lap 397.7 m.



Figure 18 – The Offset and Lap 1…4 lines on the graph.

The **Use the logger lap index as lap source** option sets the *Logger lap index* source: enable it if the dataset also has GPS laps that you want to replace.

> **Example**
>
> On the Medicina dataset, with an offset of 195.0 m and a length of 397.7 m, the wizard finds 5 crossings and 4 complete laps (47.8 s, 39.1 s, 37.8 s, 35.3 s), very close to the GPS times.

7 Circuit map and GPS trajectory

7.1 Circuit Map

The **Circuit Map (CSV Normalized)** window draws the track from a CSV file and shows:

- the **position dot** of each dataset under the cursor (in the dataset color);
- the **part of the track visible** in the graph, highlighted for every dataset in its color (the selected dataset with a thicker line);
- the **finish line star**.
- the **GPS trace** of every loaded dataset, in the dataset color (**GPS traces** checkbox);

Clicking on the map moves the graph cursor to the corresponding point.

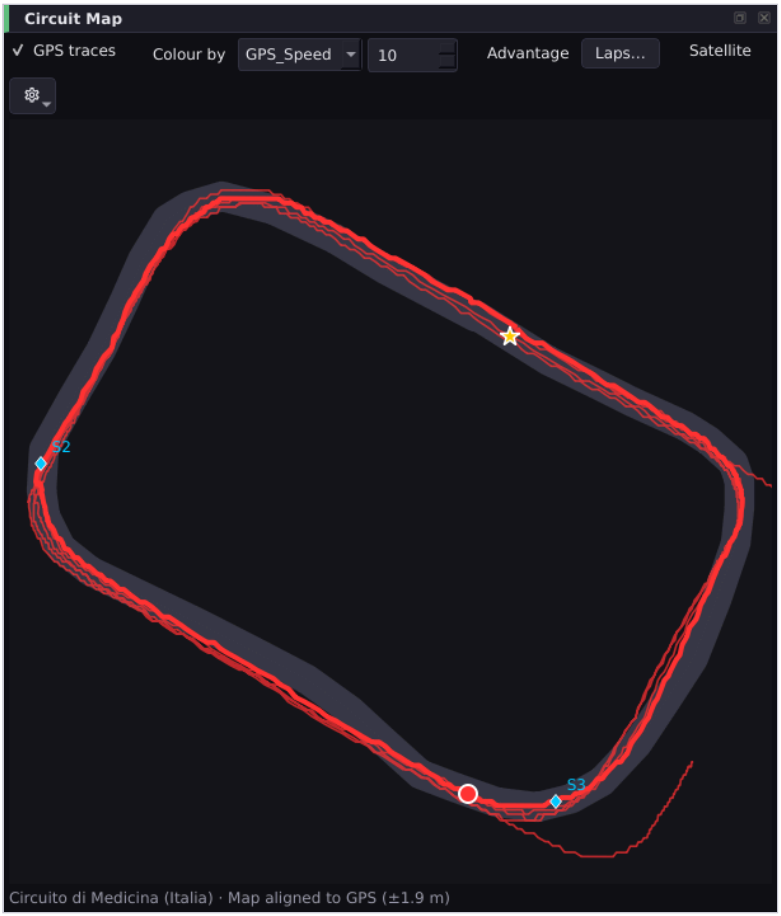


Figure 19 – Medicina map aligned to the GPS (mean deviation ±1.9 m), with the visible section highlighted.

The bottom line summarizes the status: map name, quality of the alignment to the GPS, reference lap length and position source (*GPS* or *distance*).

GPS traces. With the **GPS traces** checkbox enabled, the real GPS trajectory of every loaded dataset (valid fixes only) is drawn over the track in the dataset color: with several datasets the trajectories are overlaid and the lines driven in the different sessions can be compared. If no CSV map is loaded, the window still shows the GPS trajectories of the datasets.

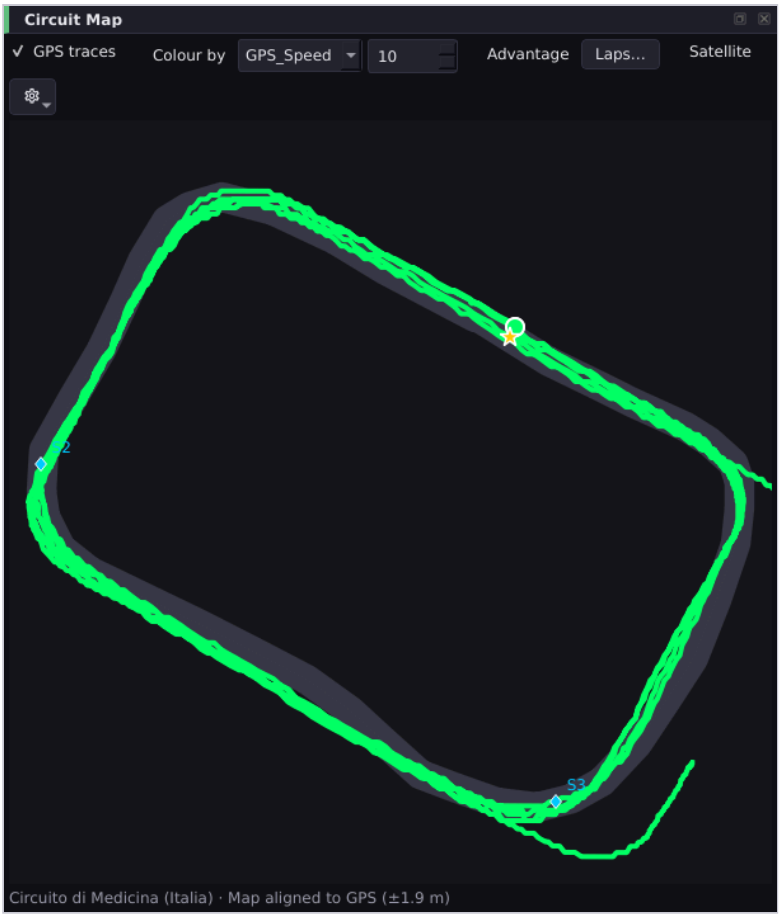


Figure 20 – GPS traces of two datasets overlaid on the map (here two copies of the same session, so they coincide).

7.2 Loading or creating the map

- **Automatic**: a `*_track.csv` file in the dataset folder is loaded when the dataset is opened.
- **Import CSV…** (or **File → Import Circuit Map (CSV)…**): loads a map from the `TrackMap` folder or another location.
- **Map from GPS**: creates the map from the reference lap of the dataset and saves it as a **georeferenced** CSV (columns `X,Y,LAT,LON`).

Hand-drawn maps (X,Y only) are **automatically aligned** to the GPS trajectory (scale, rotation, position and finish line). Without GPS, the position is calculated from the distance within the lap.

Finish offset moves the finish line along the track as a percentage of its length, for non-georeferenced maps.

Included circuits and automatic recognition. The `TrackMap` folder contains 5265 circuits from all over the world, the same as GUBELLINI DataStudio (catalog `track_maps.xml` with name, country and finish line). The maps are georeferenced (X,Y,LAT,LON columns): datasets with GPS are drawn at their real position and the finish line is the first point of the map. The name shown comes from the catalog, e.g. *Mugello Circuit (Italia)*. When a dataset is opened, the program first uses the CSV map in the dataset folder (copied by DataStudio at download); only if it is missing does it compare the GPS trajectory with the maps of `TrackMap`, limited to the circuits whose finish line is near the trajectory, and load the recognised one (notification *“Circuit recognised from the GPS trajectory”*). Where a venue has several layouts (e.g. Nürburgring GP, DTM, Nordschleife) the one matching the trajectory is chosen. New maps created with **Map from GPS** or with the Track editor and saved in `TrackMap` are always compared.

7.3 GPS Trajectory Map

Draws the actual latitude and longitude of each dataset, with the position dot at the cursor. It is useful for checking the quality of the GPS signal and for seeing the different trajectories, the pit exit and the pit entry.

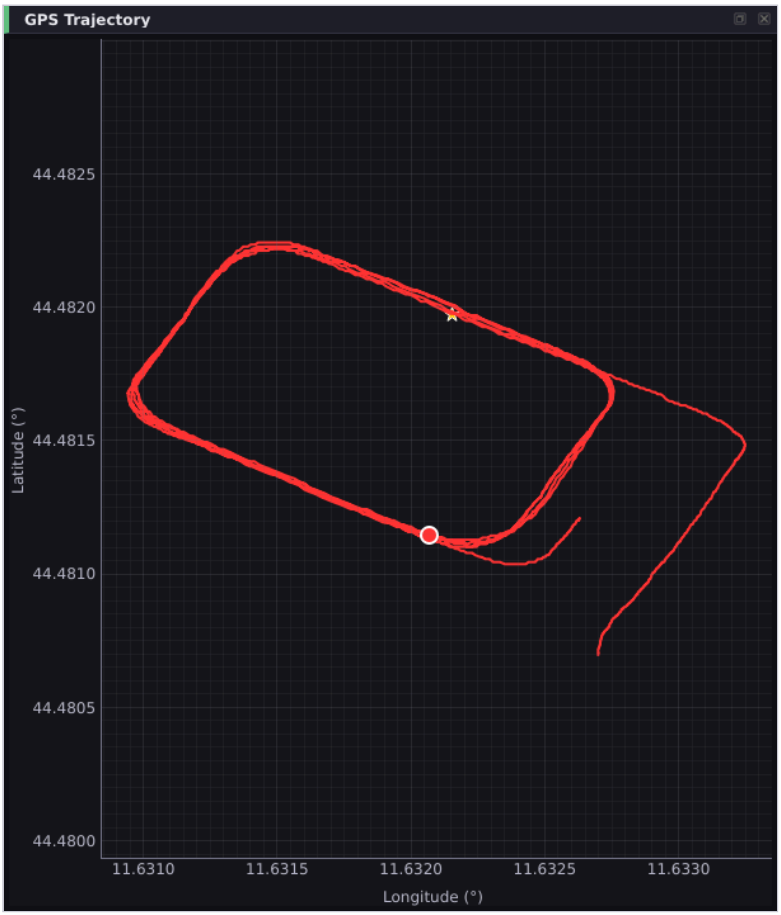


Figure 21 – GPS trajectory of the sample session.

7.4 GPS coherence

The fix is considered invalid when the GPS status is `V`, the coordinates are zero or out of range, the position stays still while the vehicle is moving, or the jumps are incompatible with the maximum acceleration. The parameters are in the **GPS coherence** tab of the Preferences (chap. 18).

8 Channel layouts

A **layout** defines which channels appear in the graph, in what order, with which color, line width, vertical band and scale. The active layout is chosen from the **Layout** box in the toolbar or with `Ctrl`+`1`…`Ctrl`+`9`.

8.1 Layout editor

Layouts → Define / Edit Layout… (`Ctrl`+`L`).

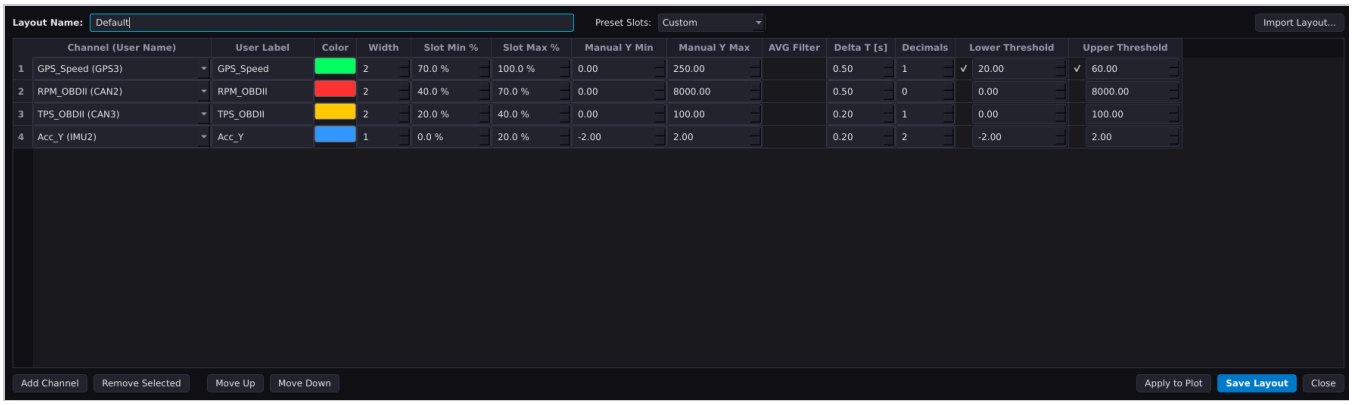


Figure 22 – The editor of the Default layout.

Column	Meaning
Channel (User Name)	Logger channel or math channel.
User Label	Displayed label. If the name comes from `Channels_details.xml` the field is locked and shows the note *«Label from Channels_details.xml of the dataset»*.
Color, Width	Line color and width.
Slot Min % / Slot Max %	Vertical band of the graph occupied by the channel (0% = bottom, 100% = top). Channels with the same band overlap.
Manual Y Min / Max	Scale limits in ManualZoomY.
AVG Filter	Moving average to reduce noise.
Lower Threshold / Upper Threshold	Lower and upper threshold of the channel: checkbox to enable it and value. They draw a horizontal reference line in the channel band.

Buttons: **Add Channel**, **Remove Selected**, **Move Up/Down**, **Apply to Plot** (preview), **Save Layout**. **Preset Slots** distributes the bands automatically; **Import Layout…** imports a layout from a file.

> **Example – Default layout**
>
> GPS_Speed at the top (band 70–100%, scale 0–250), RPM_OBDII in the 40–70% band, TPS_OBDII 20–40% and Acc_Y 0–20% with a scale of −2…+2 g: this is the layout in figure 2.

Channel thresholds

For every channel of the layout two thresholds can be enabled, **Lower Threshold** and **Upper Threshold**, with their value (in the channel unit). In all graph windows – main graph and additional

graphs – every active threshold becomes a thin (1 px) dotted horizontal line in the channel color at 50% transparency, drawn only in the band occupied by the channel. They are a visual reference, for example a minimum cornering speed or the shift RPM. A threshold outside the current scale of the channel is not drawn.

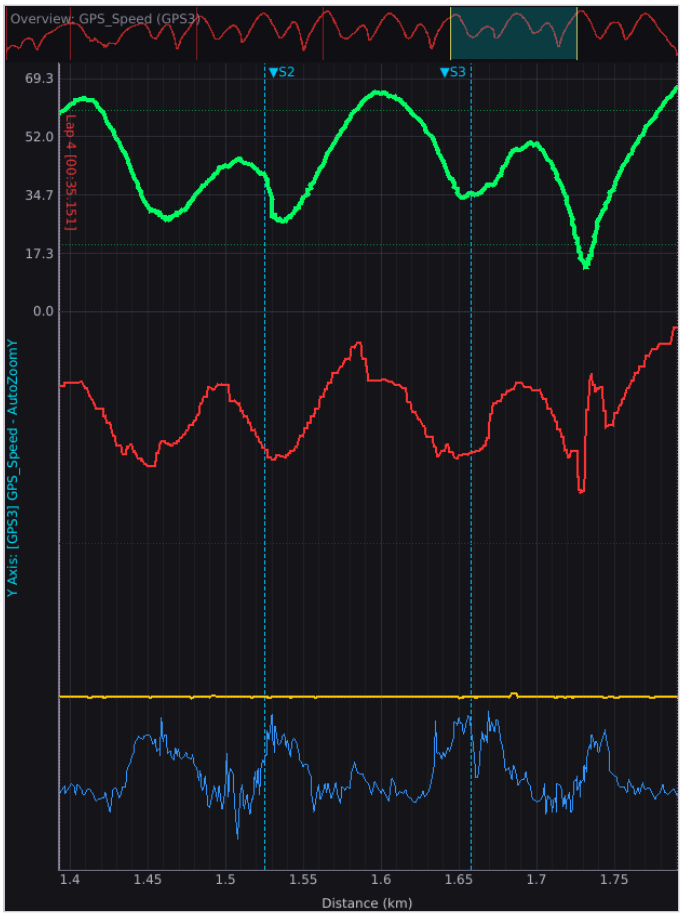


Figure 23 – GPS_Speed thresholds at 20 and 60 km/h on lap 4: the two green dotted lines in the speed band.

8.2 Saving and importing

Layouts → Save Channel Layout (`Ctrl`+`S`) and **Save Channel Layout As…** (`Ctrl`+`Alt`+`S`) save the layout in the `Layouts` folder; **Import Layout…** adds a layout received from a colleague.

9 Comparing datasets and laps

9.1 Shifting a dataset

To overlay two datasets (or two sessions), the selected one is shifted along the horizontal axis:

1. Select the dataset in the status bar and press `S` (or **Shift**).
2. Drag with the mouse, or use the wheel or the arrow keys; the shift value is shown on the graph.
3. `Enter` confirms, `Esc` cancels. `Shift`+`S` resets the shift.

Double-clicking a time in the laps window automatically aligns that lap with the left edge of the view. When switching from time base to distance base (and vice versa), the shifts are converted so that the alignment is kept.

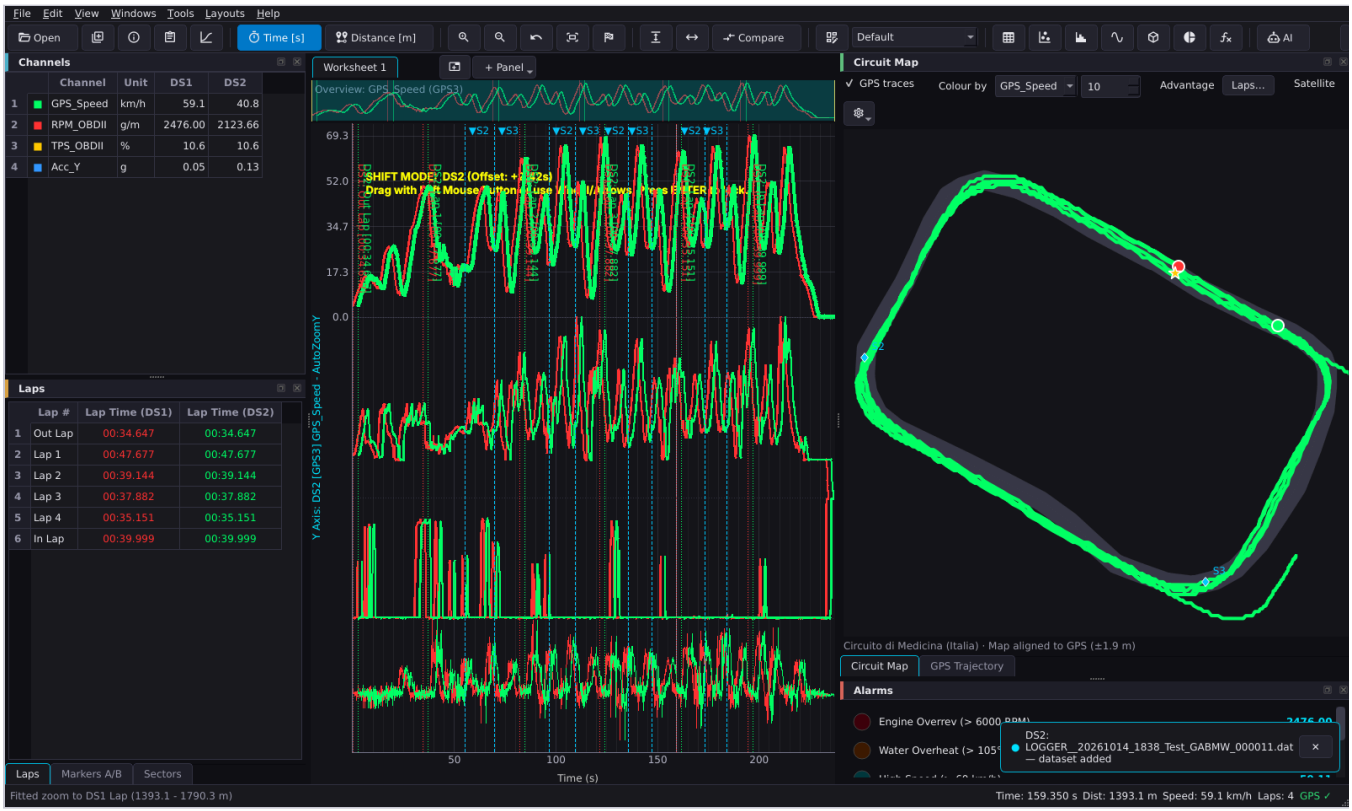


Figure 24 – Two datasets in shift mode (SHIFT MODE): DS2 is being aligned with DS1.

9.2 Lap comparison (ΔT and sectors)

Tools → Dual Dataset Comparison Mode (`Ctrl`+`D`) opens the comparison panel below the graph:

- **Reference** and **Compare**: the two laps to compare, which may also belong to the same dataset.
- **Align**: shifts the compare dataset so that the two laps start together.
- **ΔT** curve: time difference accumulated along the lap, calculated on the corrected distance (valid in both time base and distance base). If it rises, the compare lap is losing time.
- **Δ from window start**: resets ΔT to zero at the start of the displayed part, to analyze a single section.
- **Sector table**: automatic sectors on the straights or of equal length (adjustable number), with the reference and compare times and their difference; the best sectors are highlighted and the

theoretical best lap (sum of the best sectors) is calculated.



Figure 25 – Lap 4 (reference, 35.151 s) versus Lap 2 (39.144 s): final ΔT +3.993 s; the largest gap is in sector 2.

With the comparison active, the **circuit map is colored** green where the compare lap gains and red where it loses.



Figure 26 – The comparison panel in the main window.

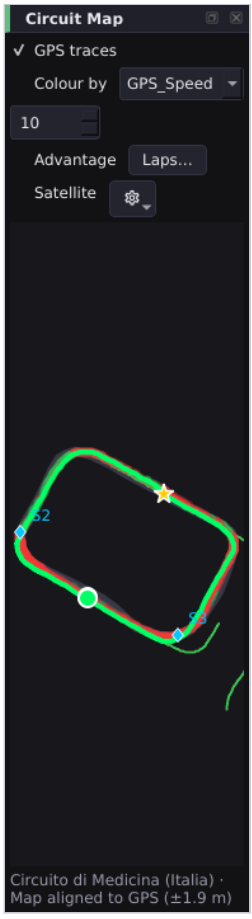


Figure 27 – Map colored by gain/loss.

Example – where time is lost

In the figure, ΔT grows mainly between 1.55 and 1.6 km and between 1.75 and 1.8 km: at those points the GPS_Speed of lap 2 is lower on corner exit. The sector table confirms it: S1 +0.332 s, S2 +2.230 s, S3 +1.431 s.

9.3 ADV Compare

Tools → ADV Compare (`Ctrl`+`Shift`+`A`) overlays many laps, even from different datasets, in a dedicated window:

- on the left you tick the laps to compare (**All** / **None**) and choose up to 8 channels;
- the laps are **normalized** to the best lap (in time or in distance);
- for each dataset a **transparent min/max band** shows the spread between the laps;
- the cursor shows the *Best*, *Max* and *Min* values for each channel;
- the configuration is saved with **Save layout** in the `ADVCompareLayouts` folder.



Figure 28 – ADV Compare with GPS_Speed, RPM_OBDII and Acc_Y for all the laps of two datasets.

10 Additional graphs and worksheets

10.1 Additional graphs

Tools → Add graph (`Ctrl`+`G`) adds up to 4 graphs below the main graph (*Graph 1, Graph 2…*), each with its own layout chosen from the box at the top. Zoom and cursor are synchronized with the main graph. The × button closes the graph. Layouts created or saved with **Save layout** or with the layout editor appear immediately in the list of every additional graph; a graph that uses the layout just saved is redrawn.

10.2 Worksheets

Worksheets are work tabs: each one remembers the main layout, the additional graphs and the comparison panel. You can prepare, for example, an "Engine" worksheet and a "Chassis" worksheet and switch between them with one click.

- New worksheet: button to the right of the tabs or `Ctrl`+`T`.
- Next / previous: `Ctrl`+`Tab` / `Ctrl`+`Shift`+`Tab`.
- Double-click the tab to rename it; use the context menu to duplicate or delete it.
- The zoom stays unchanged when switching worksheets.

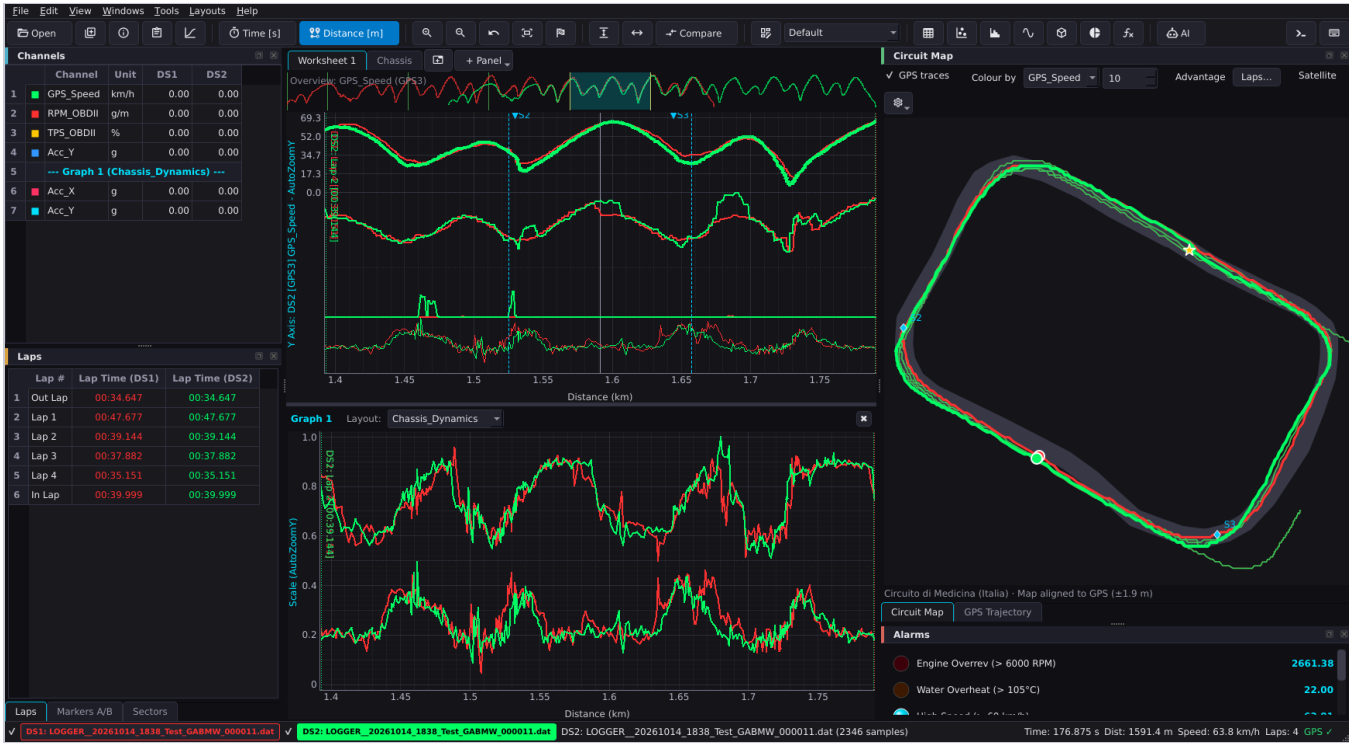


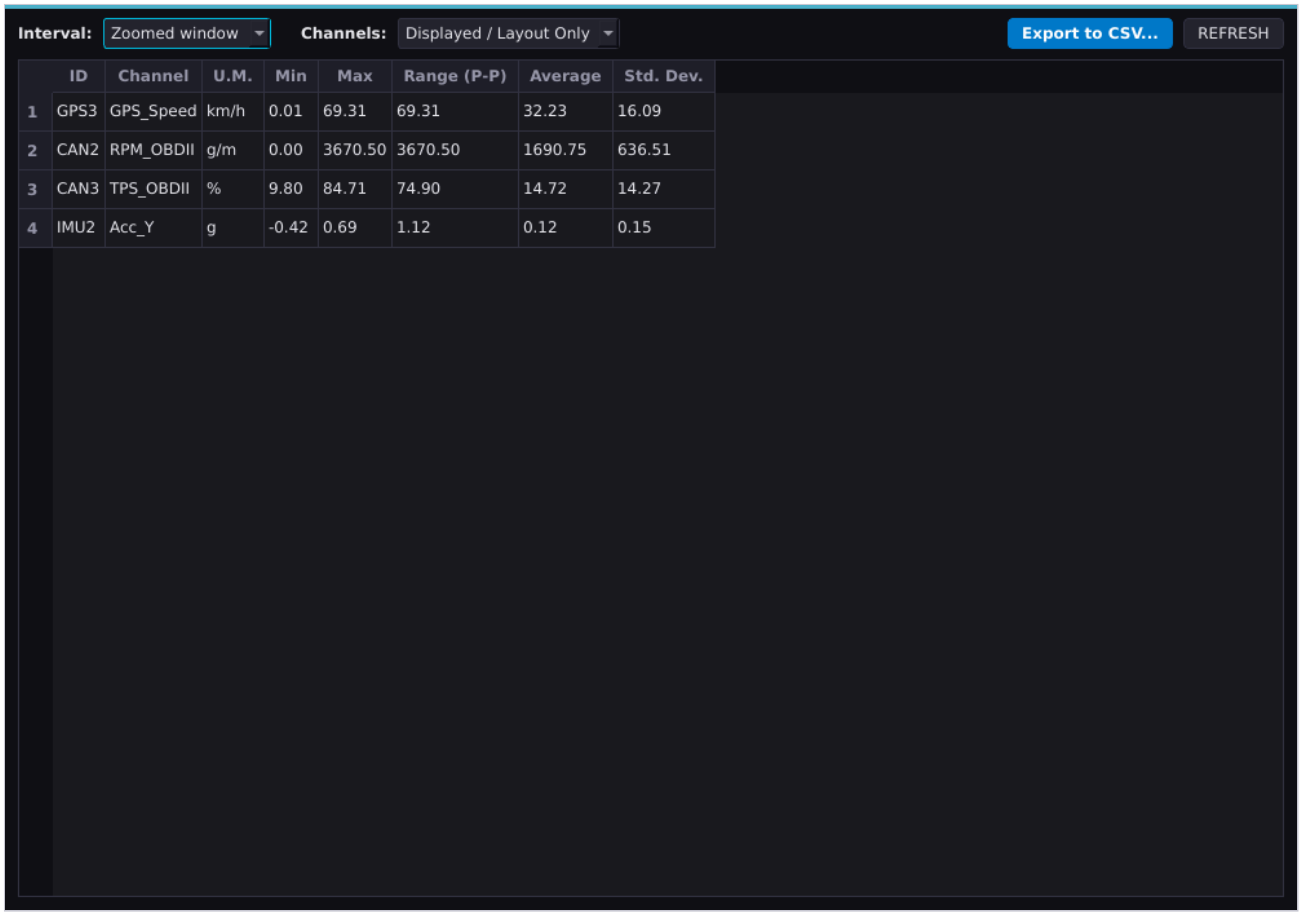
Figure 29 – "Chassis" worksheet with an additional graph using the Chassis_Dynamics layout.

11 Analysis windows

They are opened from the **Windows** menu, from the toolbar or with keys F5…F10. In all of them you can choose whether to analyze the **whole dataset** or only the **zoomed window**: to analyze a single lap, just zoom on it (L) and press **REFRESH**.

11.1 Statistics (F5)

For each channel: minimum, maximum, range, average and standard deviation. **Channels** limits the table to the layout channels or includes all of them; **Export to CSV…** saves the table.



	ID	Channel	U.M.	Min	Max	Range (P-P)	Average	Std. Dev.
1	GPS3	GPS_Speed	km/h	0.01	69.31	69.31	32.23	16.09
2	CAN2	RPM_OBDII	g/m	0.00	3670.50	3670.50	1690.75	636.51
3	CAN3	TPS_OBDII	%	9.80	84.71	74.90	14.72	14.27
4	IMU2	Acc_Y	g	-0.42	0.69	1.12	0.12	0.15

Figure 30 – Statistics of the zoomed window.

11.2 XY scatter plot (F6)

One channel plotted against another, with optional coloring by a third channel (**Gradient Ch**) and a connecting line. Typical examples: Acc_X versus Acc_Y (G-G diagram), RPM versus speed (gear ratios), latitude versus longitude.

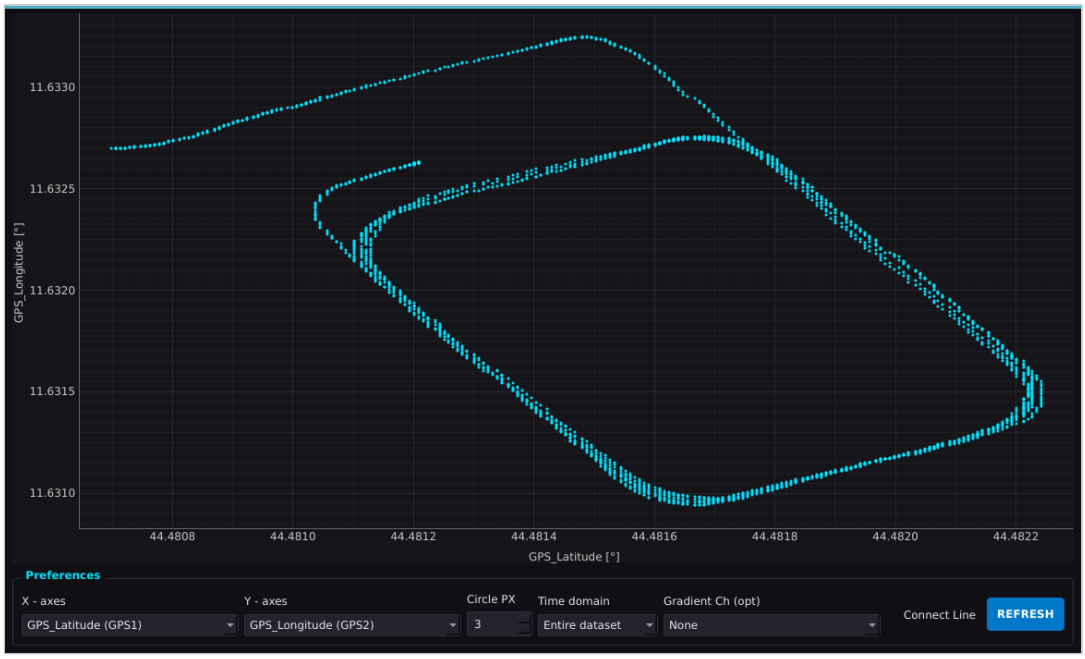


Figure 31 – GPS_Latitude / GPS_Longitude scatter plot.

11.3 Histogram (F7)

Distribution of the values of a channel over the chosen number of classes (**BINS**): for example, how much time is spent at each speed or at each throttle opening.

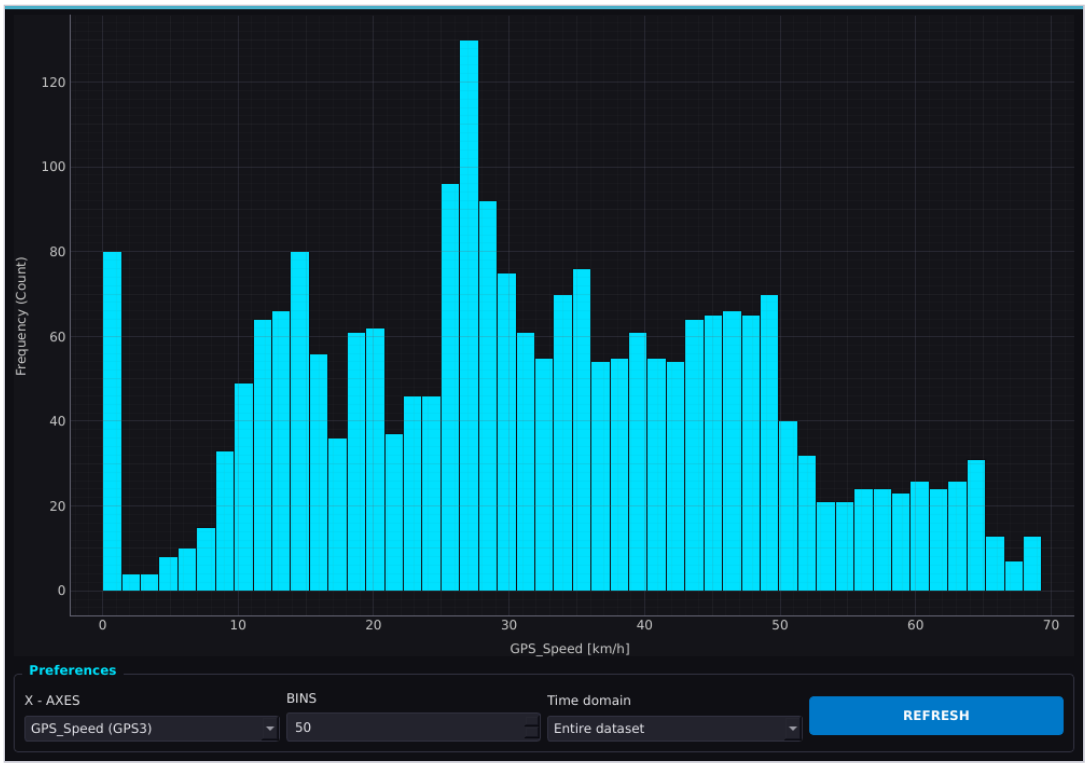


Figure 32 – Histogram of GPS_Speed over 50 classes.

11.4 FFT spectral analysis (F8)

Frequency spectrum of a channel, with a Hanning/Hamming/rectangular window and a linear or dB scale. It is used to identify vibrations and periodic oscillations (suspension, chassis, engine).

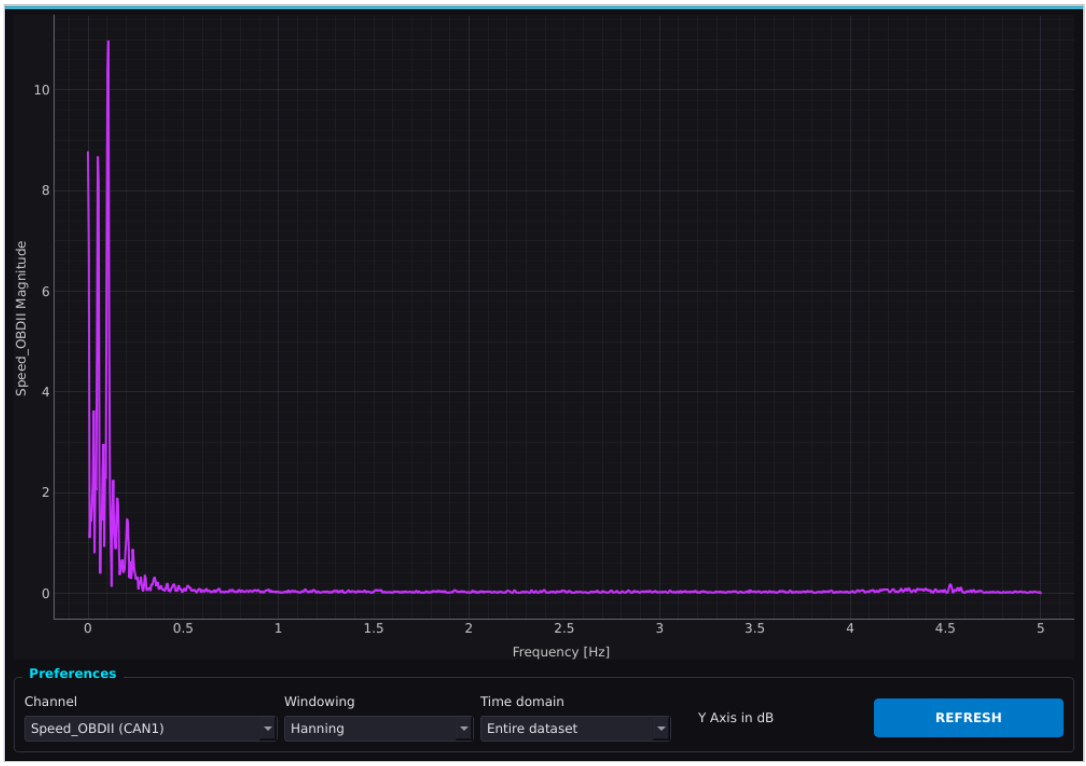


Figure 33 – FFT spectrum of the speed.

11.5 3D XYZ analysis (F9)

Point cloud with three channels on the three axes, which can be rotated with the mouse. Example: Acc_X, Acc_Y and Acc_Z to see the acceleration envelope.

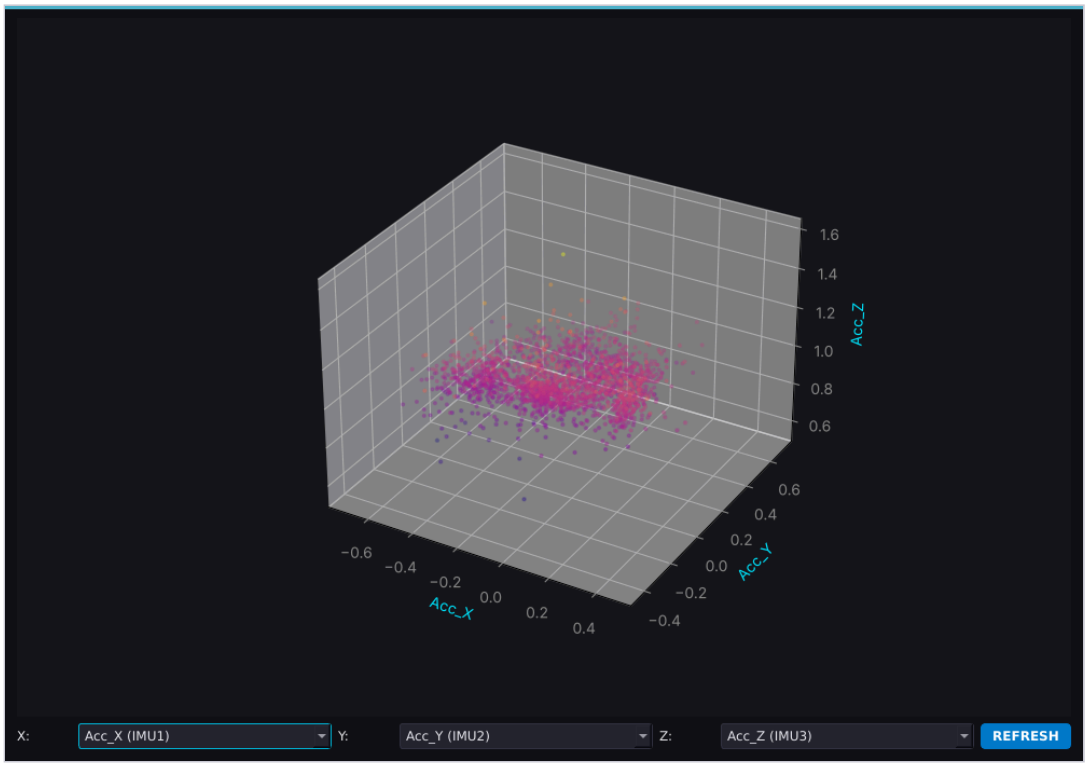


Figure 34 – IMU accelerations in 3D.

11.6 Pie chart (F10)

Percentage of time spent in each value interval of a channel, with the chosen number of sectors.

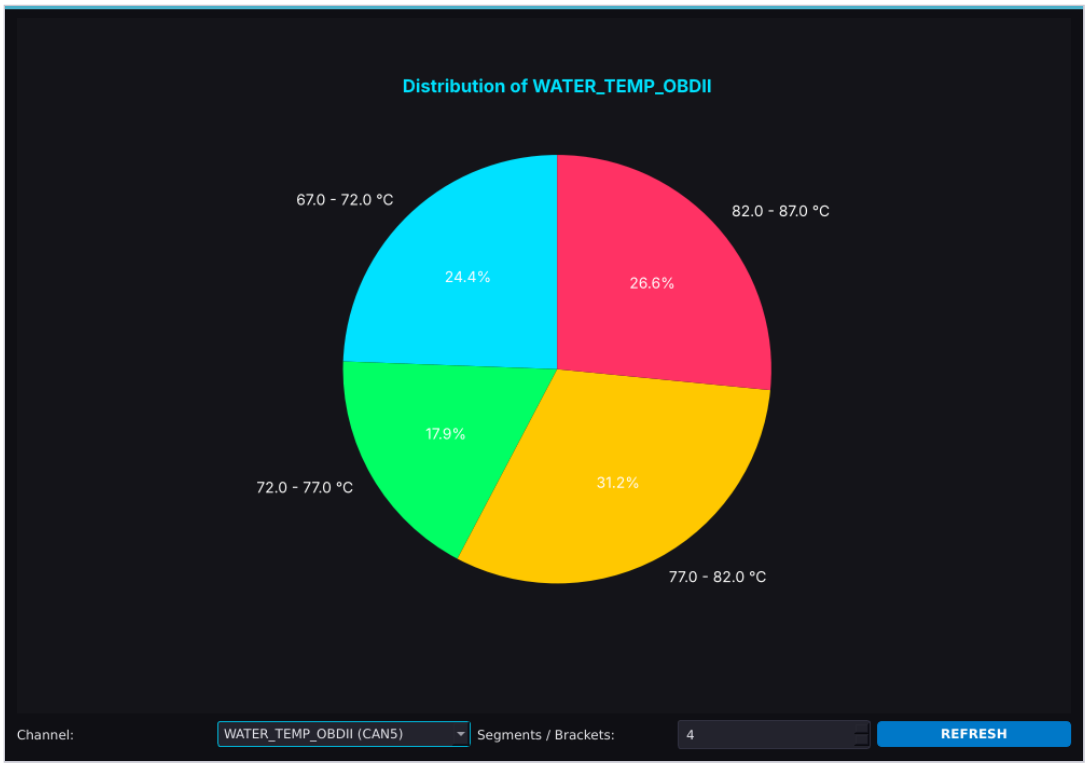


Figure 35 – Distribution of the water temperature (WATER_TEMP_OBDII) in 4 intervals.

12 Math channels and lookup tables

12.1 Math channels manager

Tools → Math Channels Manager… (`Ctrl` + `M`) creates new calculated channels from the recorded ones. A math channel can be used like any other channel: in layouts, statistics, graphs and other math channels. The channels are saved automatically and reloaded in the next session.

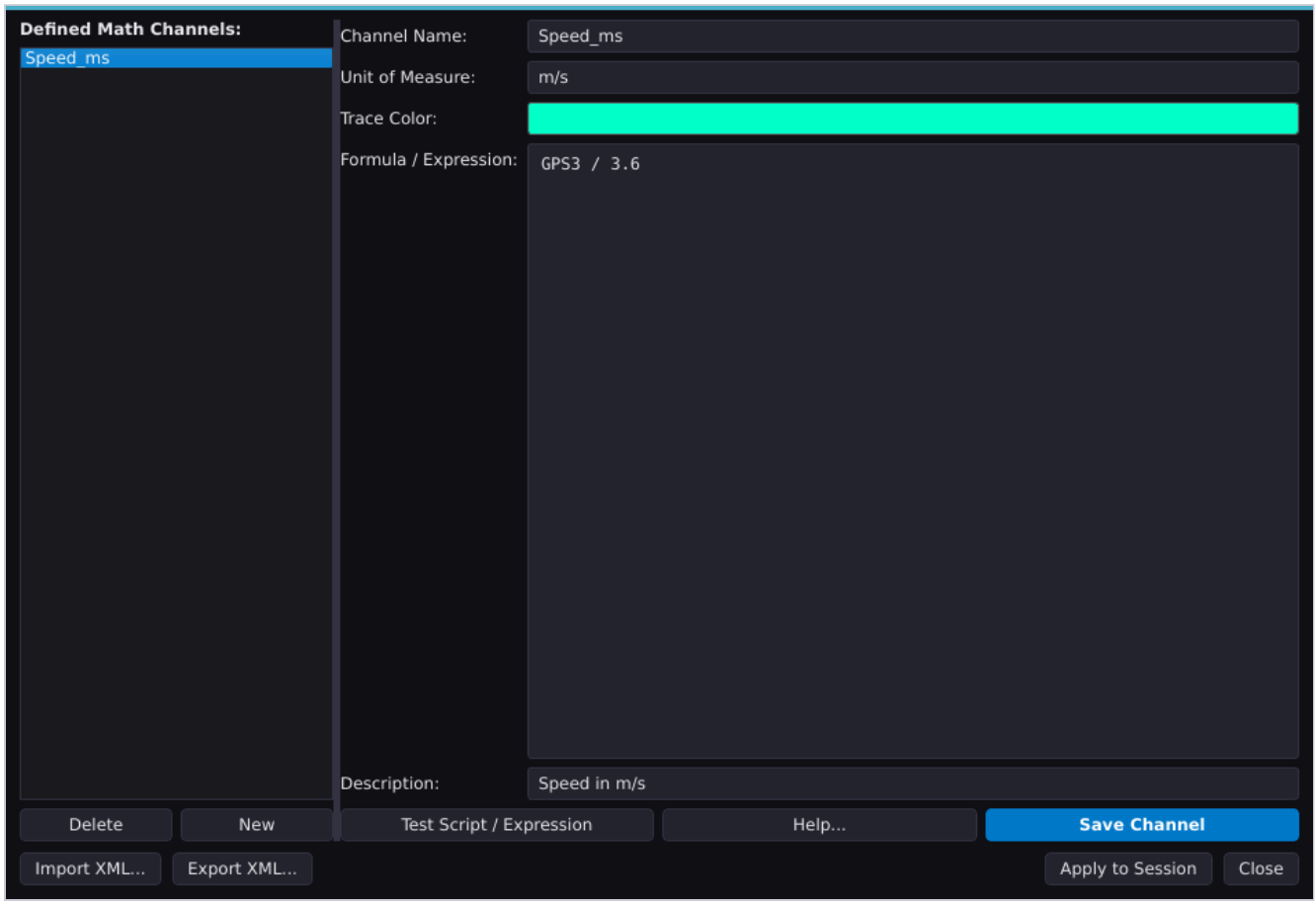


Figure 36 – The Speed_ms channel (speed in m/s) defined as GPS3 / 3.6.

1. Press **New** and enter the name, unit of measurement, color and description.
2. Write the formula in **Formula / Expression**, using the channel names (e.g. `GPS3` or the user label).
3. **Test Script / Expression** checks the formula; **Save Channel** saves it; **Apply to Session** calculates the channel on the open datasets.

Formulas can be a simple expression or a multi-line script with `if`, loops and `return`. Mathematical functions are available (`sqrt`, `sin`, `cos`, `exp`, `abs`…) as well as telemetry functions such as `diff` (derivative), `integ` (integral), `smooth` (filter) and `where` (condition), plus the `time` and `distance` vectors.

Channel	Formula
Speed in m/s	`GPS3 / 3.6`
Longitudinal acceleration [g]	`diff(GPS3 / 3.6) / 9.81`
Full throttle (0/1)	`where(TPS_OBDII > 90, 1, 0)`
Combined acceleration [g]	`sqrt(Acc_X**2 + Acc_Y**2)`

Help… opens the complete guide to the calculation engine, with an index, search and examples; **Import XML…** / **Export XML…** exchange channels with other users.

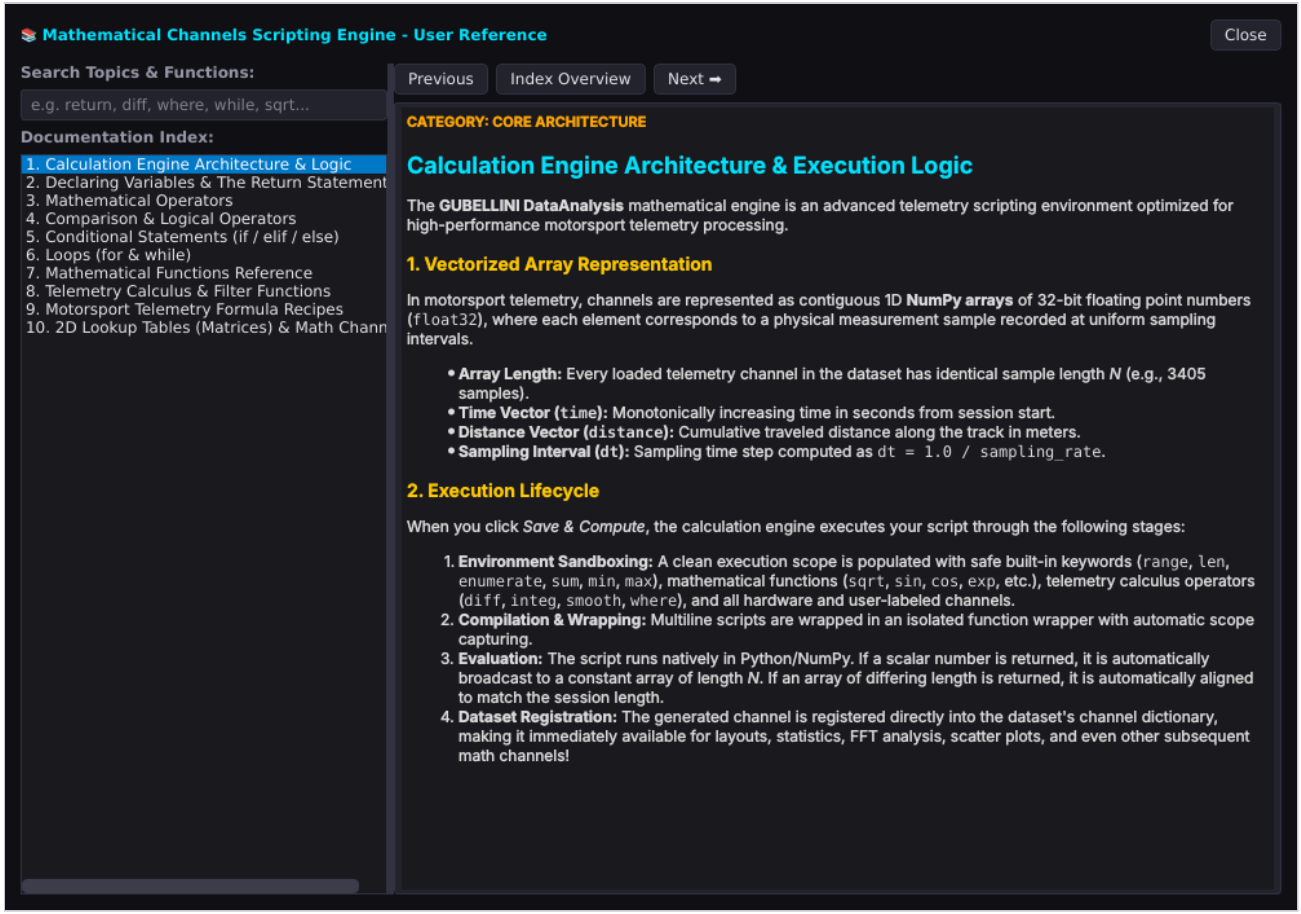


Figure 37 – The math channels reference guide.

12.2 Lookup tables

Tools → Lookup Table manages 2D tables (rows × columns) that can be called from math channels, for example gear ratios or a torque map. Values can be pasted from Excel with `Ctrl`+`V`.

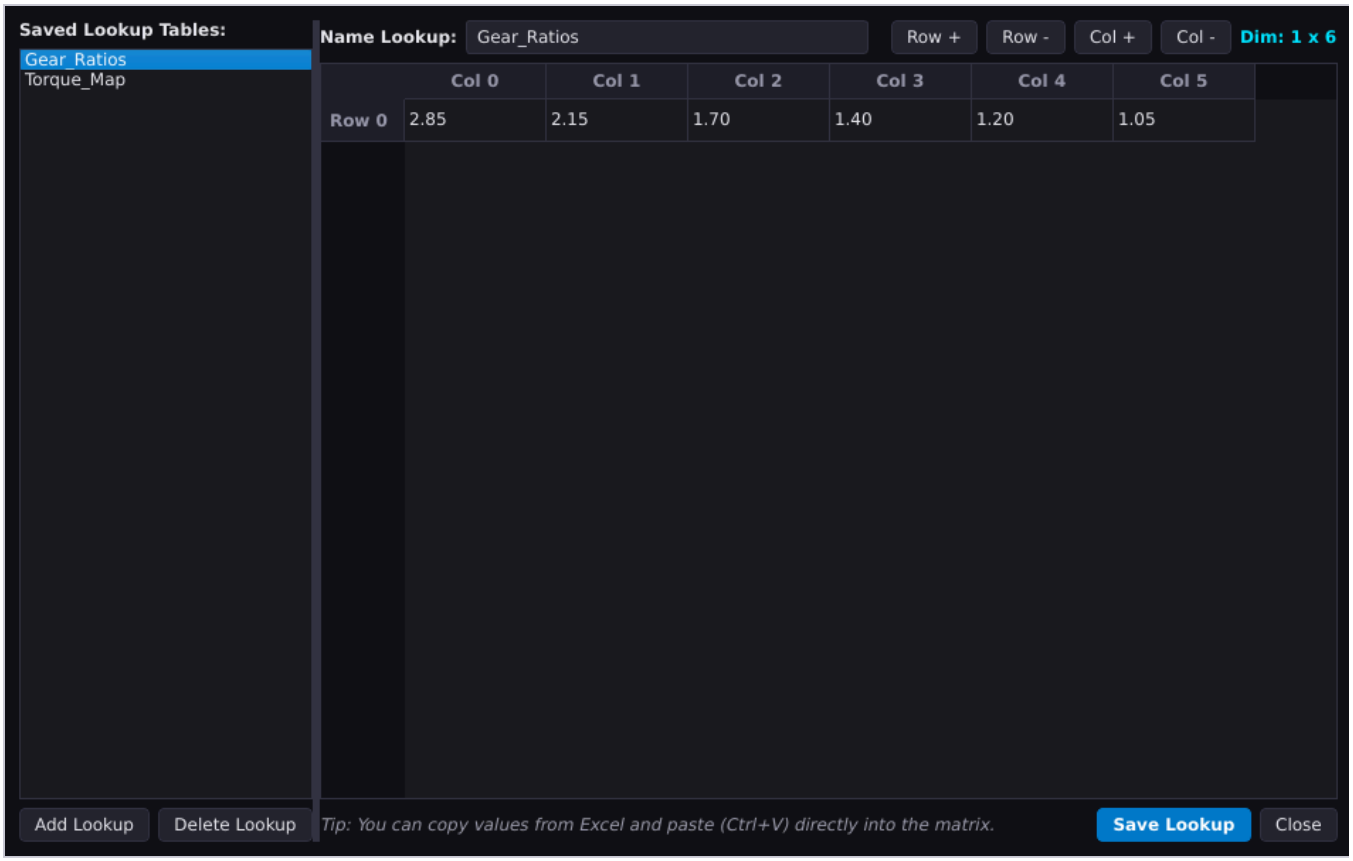


Figure 38 – The Gear_Ratios table with the gear ratios.

13 Track editor

Tools → Track editor creates a circuit map by tracing it over an image (aerial photo, screenshot of an online map) when no GPS lap is available.

1. **Open image** opens an image, or **Paste image (Ctrl+V)** pastes it from the clipboard.
2. **Start drawing**: click along the track starting from the finish line, in the direction of travel. **Undo** removes the last point.
3. **Stop drawing** closes the track; **Save track** saves it as CSV in the `TrackMap` folder.

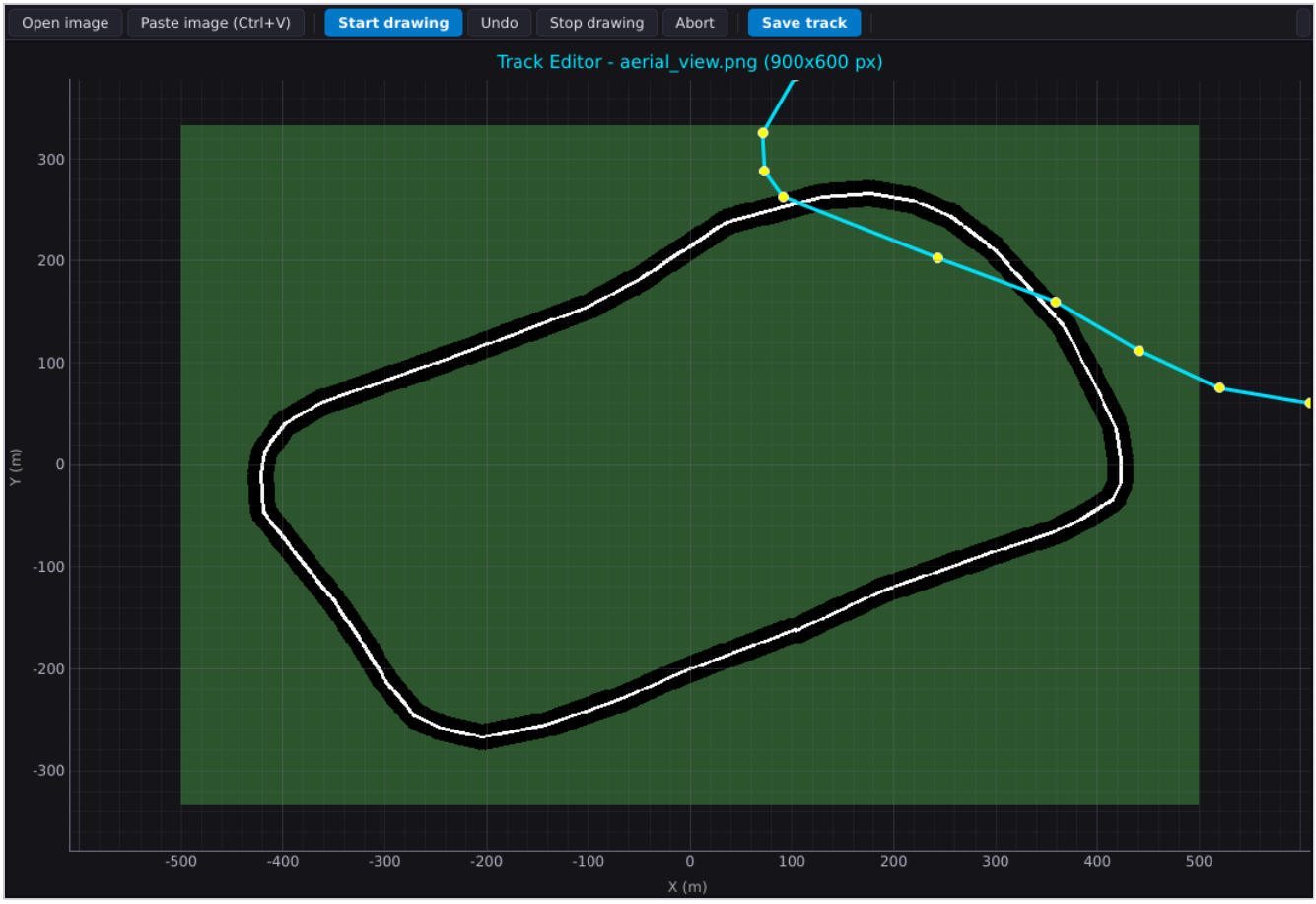


Figure 39 – The Track editor while tracing a track over an aerial image.

> **Automatic alignment**
>
> A hand-drawn map loaded with a dataset that has GPS is automatically aligned to the actual trajectory; if the dataset has no GPS, the position is calculated from the distance within the lap.

14 Onboard video

14.1 Synchronizing a video

Tools → Add video (`Ctrl`+`Shift`+`V`) opens the video window.

1. **Load Video…** and choose the file (mp4, avi, mov…).
2. Press **Synchronize**: the calibration bar appears.
3. Using the slider, bring the video to a recognizable instant (e.g. crossing the finish line) and the graph cursor to the same instant, then press `Enter`. The calculated offset (*Offset*) is shown.
4. From then on, when you move the cursor on the graphs, the video shows the corresponding frame. With **Real Time** enabled the video follows the cursor instant by instant; when disabled, it updates when the cursor stops (smoother on less powerful PCs).

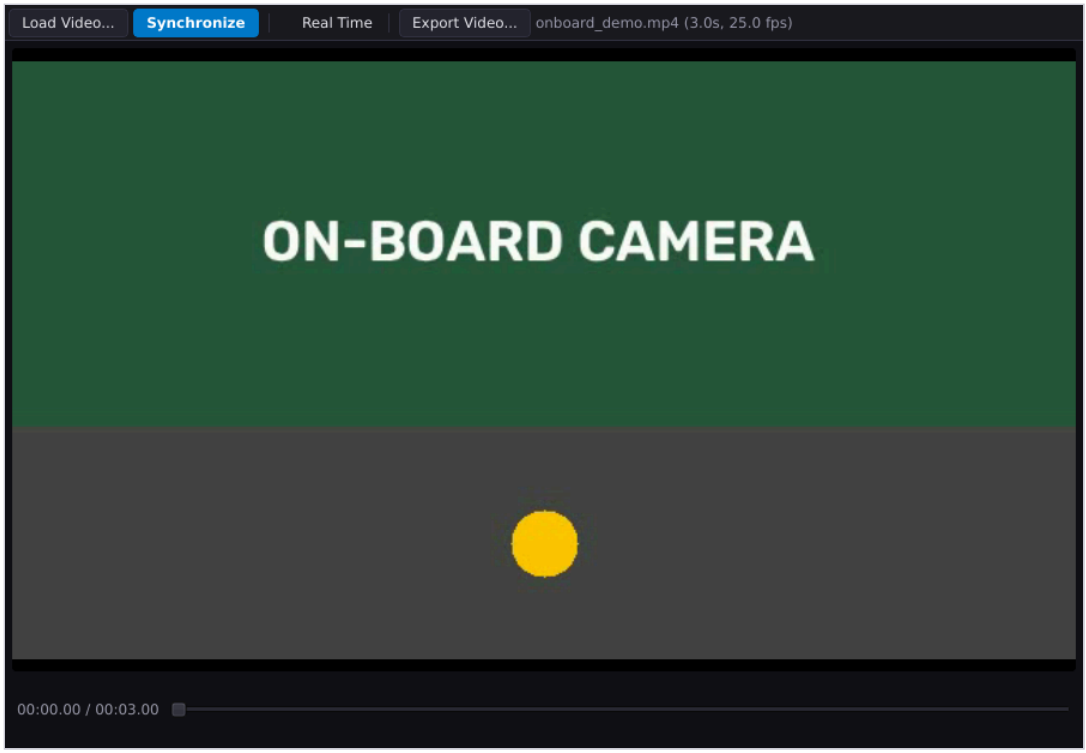


Figure 40 – The video window with a synchronized onboard clip.

14.2 Exporting the video with gauges

Export Video… creates a new clip with up to 6 overlaid gauges: rev counters, vertical bars, digital text. For each one you choose the channel, type, minimum/maximum, color, transparency and size; the position of the gauges is chosen at the top. The configuration can be saved and imported again. The video synchronization (§14.1) also drives the timer of the info box.

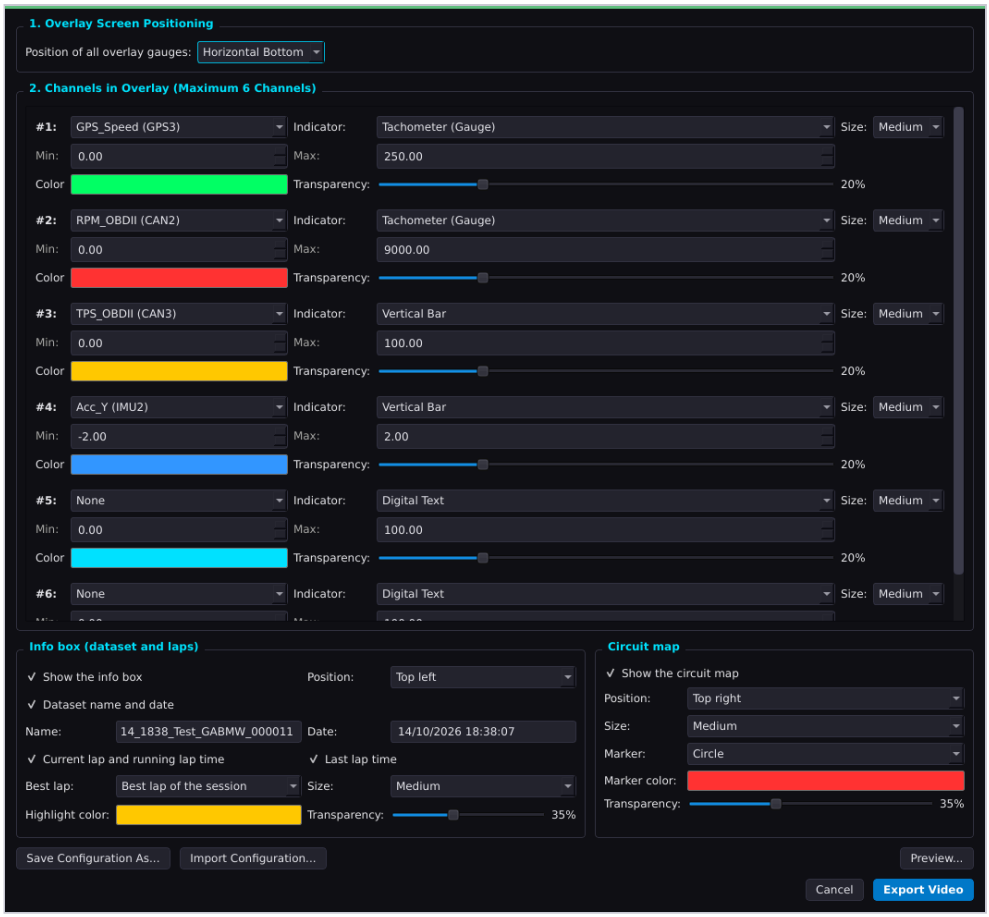


Figure 41 – Video export configuration with speed and RPM as dial gauges, TPS and Acc_Y as bars.

Info box

In the **Info box (dataset and laps)** group you can add a box with the session data to the video, configured before the export:

- **Dataset name and date**: taken from the open dataset and editable (e.g. "Medicina test");
- **Current lap and running lap time**: the lap number and a timer running **in sync with the video** (hundredths of a second) that restarts at every finish line crossing;
- **Last lap time** of the last completed lap, in **purple** when it is the best;
- **Best lap**: **Best lap of the session**, **Best lap so far** (updated as the laps are completed), **Full list of laps** (current lap highlighted, best lap in purple) or **Not shown**;
- **Position** (one of the four corners), **Size**, **Highlight color** and background **Transparency**.

Preview... shows one frame with the current settings, at the position shown in the video window. The info box settings are saved with the configuration. With the info box enabled the video can be exported even without channels.

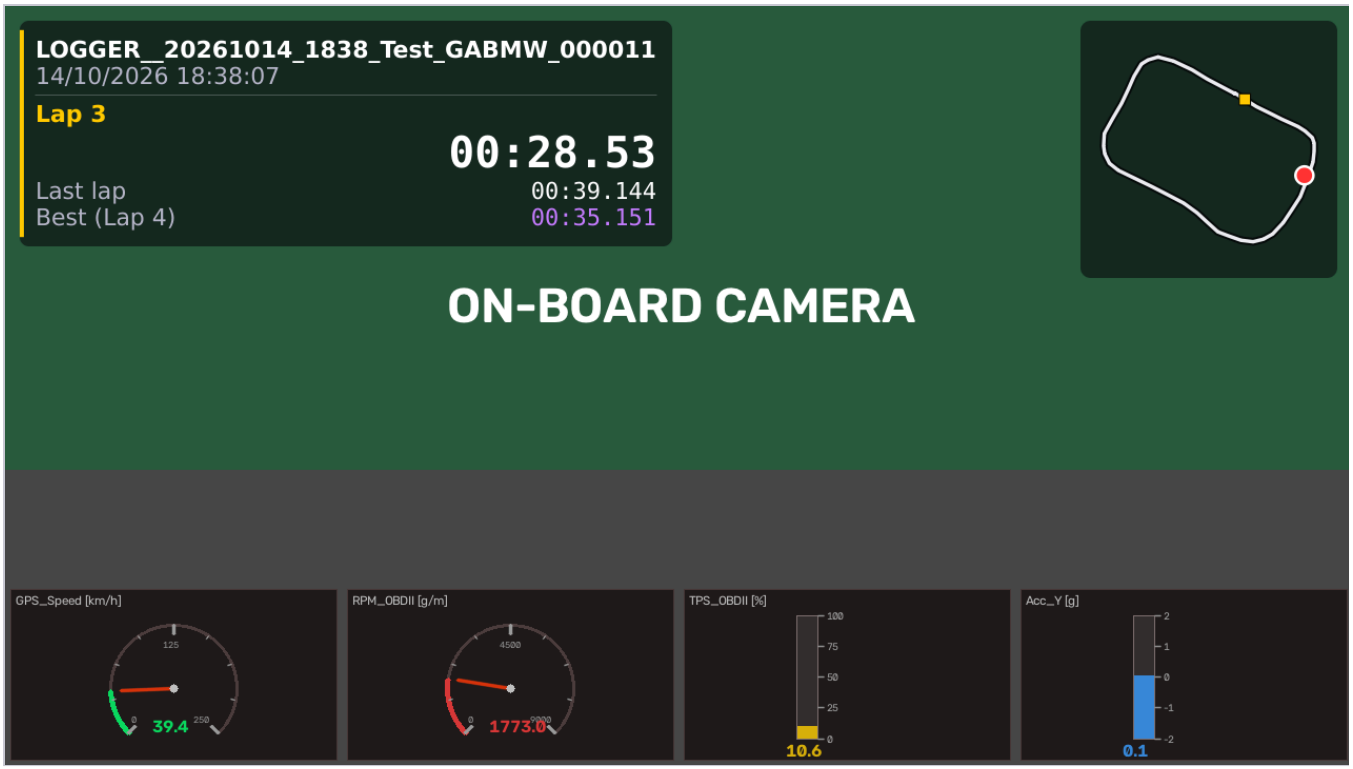


Figure 42 – Exported frame: channel gauges at the bottom and the info box with lap 3 running.

Circuit map

The **Circuit map** group adds to the video a box with the track and a **marker** showing the point of the circuit where the vehicle is. The marker is synchronized with the other overlays (gauges and info box) because it uses the same video synchronization.

- **Show the circuit map**: enables the box (available when the circuit map is loaded in the Circuit Map window or the dataset has GPS; without a CSV map the GPS trajectory is used);
- **Position** (four corners) and **Size**;
- **Marker**: **Circle** or **Cross**, with **Marker color** (the dataset color by default);
- background **Transparency** of the box.

The finish line is shown by a small yellow square. The settings are saved with the configuration and can be checked with **Preview...**.

15 Virtual LEDs, alarms and export

15.1 Virtual LEDs and alarms

The **Virtual LEDs / Alarms** window shows one LED for each alarm, lit when the condition is true at the cursor, together with the current value of the channel.

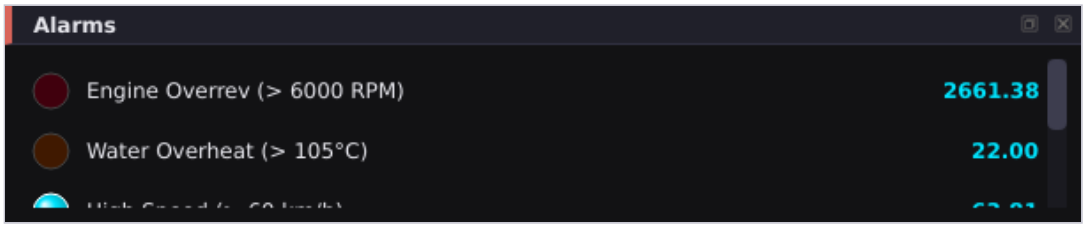


Figure 43 – Alarms at the cursor: High Speed is lit (63.81 km/h > 60).

Configure Alarms… opens the alarm table: for each alarm you set whether it is active, the channel, the condition (`>`, `<`, `=`, `between` two thresholds), the thresholds, the LED color and the description.



Figure 44 – Alarm configuration: over-rev, water temperature, speed, lateral acceleration.

15.2 CSV export

File → Export Displayed Data to CSV… (`Ctrl`+`E`) saves the data to a CSV file with units of measurement, which can be opened with Excel. You choose the dataset, the interval (zoomed window or whole dataset), the channels (displayed or all), the field separator and the decimal separator.

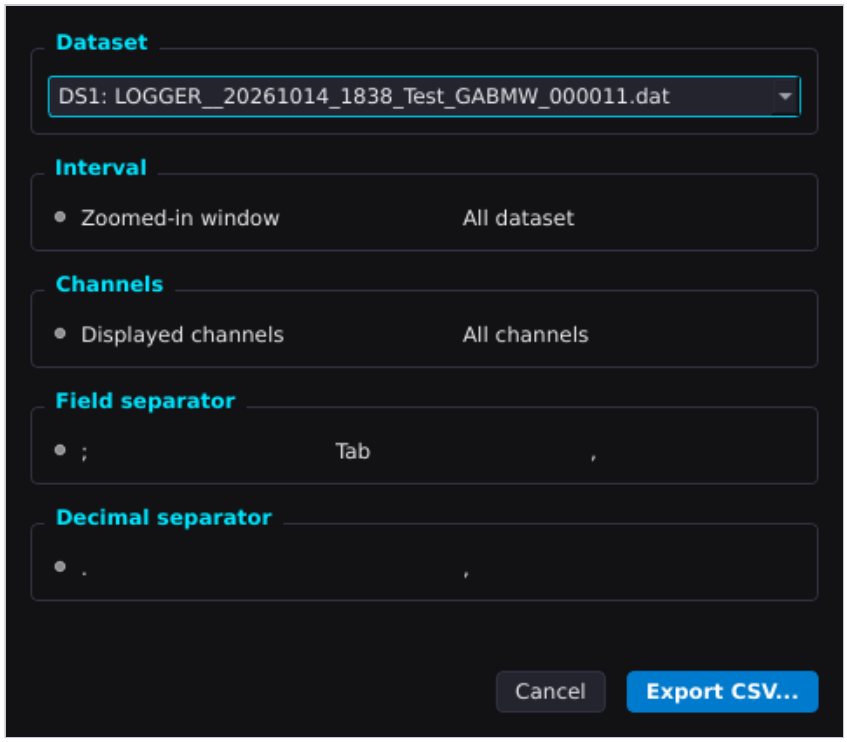


Figure 45 – CSV export options.

Choosing the separators for Excel

With an English-language Excel use the field separator , and the decimal separator . ; with regional settings that use the decimal comma (e.g. Italian, German, French) use the field separator ; and the decimal separator , .

16 AI Assistant

The AI Assistant answers questions about the loaded data, compares laps, looks for anomalies, suggests what to look at and generates reports. The numbers are calculated by the program (statistics, comparisons, braking zones, events, correlations); the model interprets and explains them.

> **With your own key, at your own expense**
>
> The assistant uses **your account** with the artificial intelligence provider you choose. Gubellini Electronics does not act as an intermediary: requests go directly from the PC to the provider, which charges costs based on usage. Alternatively, you can use a free local model (Ollama or LM Studio) that also works without internet.

16.1 Setup

1. **Tools → AI Assistant Setup…** (or the ⚙ button in the assistant window).
2. Choose the provider on the left: step-by-step instructions with links to the right pages appear on the right.
3. **Get a key…** opens the provider's website; paste the key into the **API key** field.
4. **Load models** and choose a model that supports *tool calling*.
5. **Test connection**; when «Connection OK» appears, press **Save**.

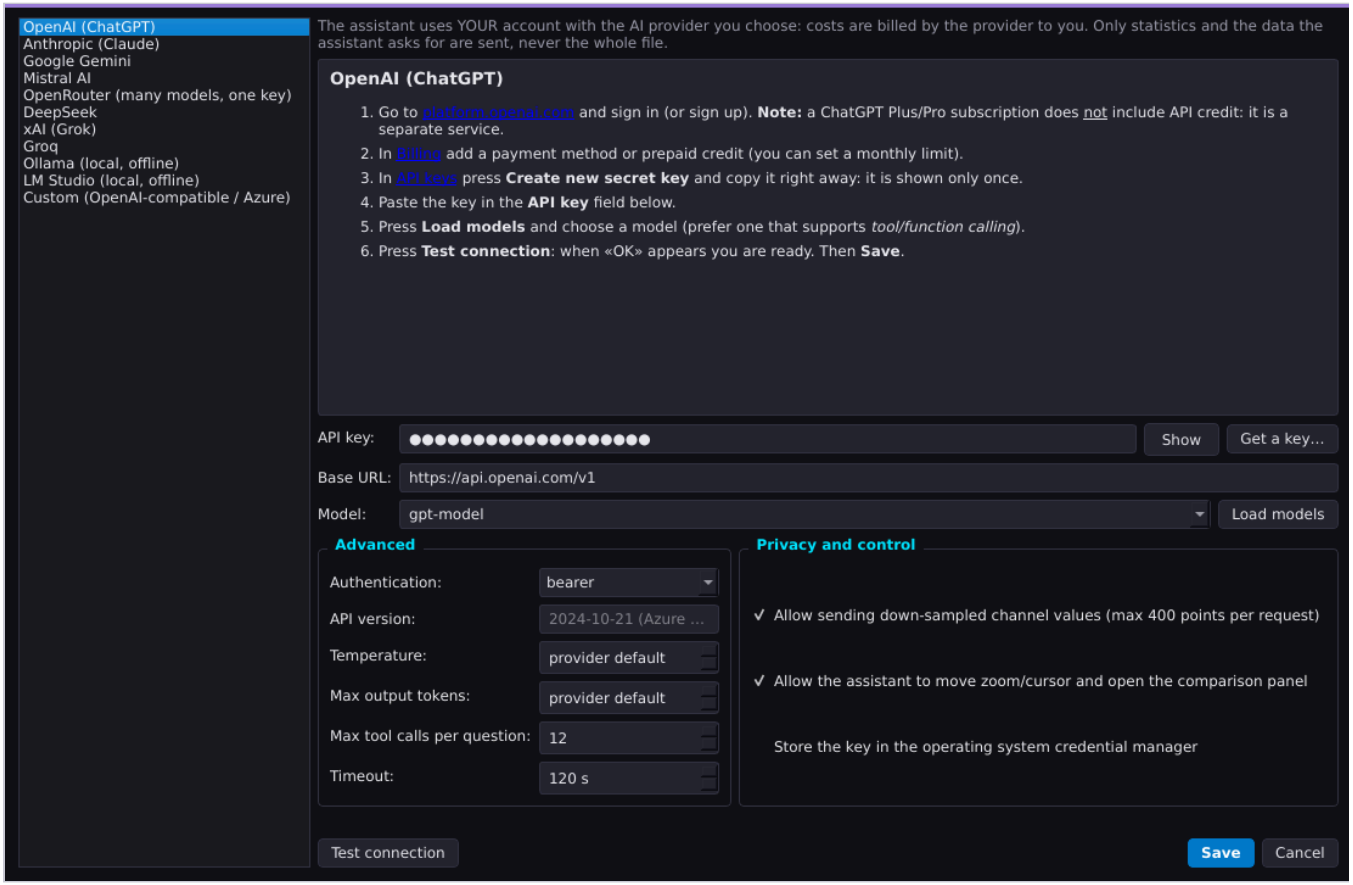


Figure 46 – AI assistant setup with the instructions for OpenAI.

Provider	Notes
OpenAI (ChatGPT)	Simplest procedure. A ChatGPT Plus/Pro subscription does not include API credit.
Anthropic (Claude)	Excellent for long analyses and reports. Credit from the Anthropic Console.
Google Gemini	Has a free tier with limits; for confidential data enable billing.
Mistral, DeepSeek, xAI, Groq	Other compatible cloud providers.
OpenRouter	A single key for models from many providers.
Ollama / LM Studio	Free, offline, the data never leaves the PC. They require a powerful PC.
Custom (OpenAI-compatible / Azure)	Corporate contracts or OpenAI-compatible services (base URL, authentication, API version).

16.2 Privacy and control

- The recording file is **never sent in full**: the program only sends the statistics and tables requested by the assistant.
- **Allow sending down-sampled channel values**: when disabled, the assistant never receives series of values (max 400 points per request when enabled).
- **Allow the assistant to move zoom/cursor**: when disabled, the assistant can only answer.
- Math channels proposed by the assistant are created only after confirmation.
- The key is stored in the Windows credential manager, if available.

16.3 Using the assistant

Tools → AI Assistant (`Ctrl`+`J`). Type the question and press `Enter` (`Shift`+`Enter` starts a new line). **New chat** clears the conversation. The token usage of the chat is shown below the text box.



Figure 47 – Example answer: sector-by-sector comparison between lap 4 and lap 1.

Example questions:

- «Summarize the session and tell me which is the best lap»
- «Where am I losing time in lap 2 compared to the best lap?»
- «Compare the best lap of DS1 with the best lap of DS2 and open the comparison»
- «Were there any over-revs above 7000 rpm? In which laps?»
- «Generate a report of the session with suggestions for improvement»

16.4 Error messages

Message	Solution
HTTP 401 / invalid key	Copy the key again without spaces, or create a new one.
HTTP 429 / credit exhausted	Add credit or wait for the free limits to reset.
HTTP 503 / model overloaded	The program retries automatically; then use **Retry** or **Choose another model**.
HTTP 404 / model not found	**Load models** and choose a model from the list.
Connection error	Check the internet connection, proxy or firewall; for Ollama/LM Studio check that the program is running.

17 Analysis report

File → Generate Analysis Report… (`Ctrl`+`R`) creates an HTML report, which can be opened with any browser and printed to PDF, containing:

- the Gubellini Electronics logo and the session data (file, date, duration, laps, best lap, percentage of GPS fix);
- a graph and a table of the lap times;
- a lap comparison with the ΔT curve and the sector table;
- braking zones and channel statistics;
- the AI assistant's summary, if the report was requested from the assistant.

Reports are saved in the `Reports` folder. The logo can be replaced in **Preferences → Report logo**.



Medicina test – Report

GUBELLINI DataAnalysis · 2026-11-13 15:30

Session

	File	Date	Duration	Full laps	Best lap	GPS fix
DS1	LOGGER__20261014_1838_Test_GABMW_000011.dat	14/10/2026-18:38.07	234 s	4	00:35.151	100%

Reference lap length: 398 m (distance correction: gps).

Analysis

Summary

- Best lap: **lap 4** (00:35.151)
- Main margin in sector 2

Lap times

DS1 – LOGGER__20261014_1838_Test_GABMW_000011.dat

Lap	Type	Time	Distance [m]	Max speed	Avg speed
Out Lap	out	00:34.647	200	49.9	20.7
Lap 1	full	00:47.677	394	57.5	29.7
Lap 2	full	00:39.144	402	65.6	36.9
Lap 3	full	00:37.882	396	68.8	37.7
Lap 4	full	00:35.151	396	65.4	40.5
In Lap	in	00:39.999	312	69.3	28.2

Lap comparison

DS1 Lap 4 (00:35.151) vs DS1 Lap 1 (00:47.677)

Figure 48 – First page of the report of the sample dataset.

18 Preferences

Edit → Preferences (`Ctrl`+`,`).

18.1 General, channels and laps

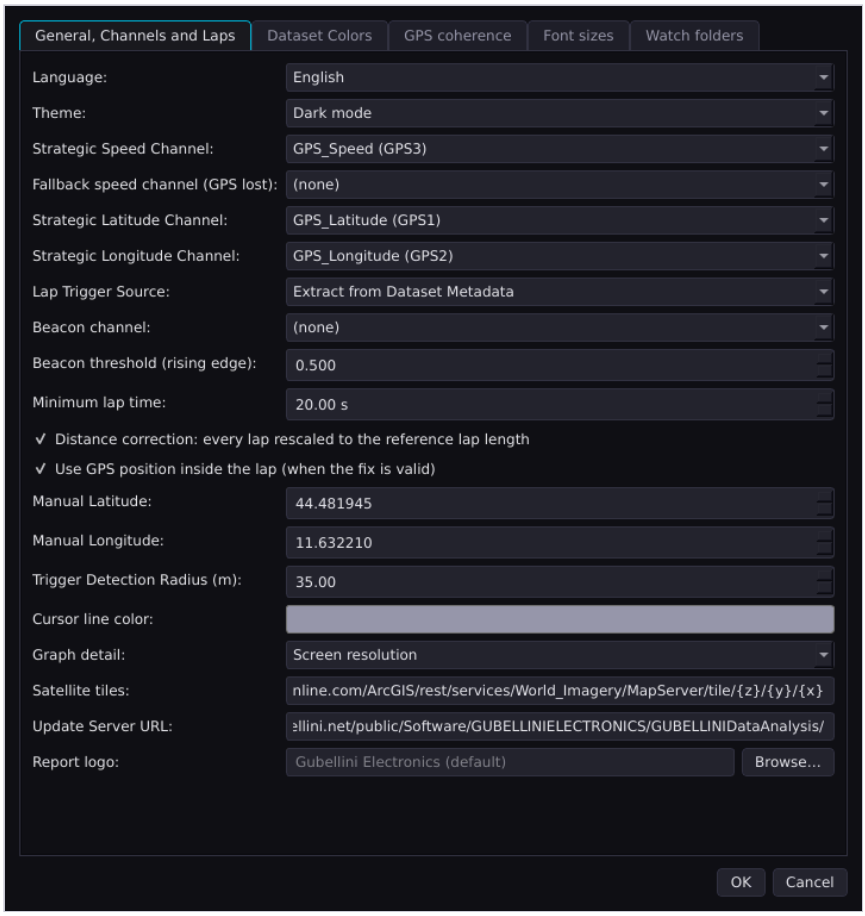


Figure 49 – The General, Channels and Laps tab.

Setting	Meaning
Language, Theme	Interface language and dark/light theme.
Strategic Speed Channel	Speed channel used for distance and overview (e.g. GPS_Speed).
Fallback speed channel (GPS lost)	Alternative speed (e.g. wheel speed from CAN) used when the GPS is not valid.
Strategic Latitude / Longitude Channel	GPS coordinate channels.
Lap Trigger Source	Source of the laps (§6.2).
Beacon channel, Beacon threshold	Channel and rising-edge threshold for beacon detection.
Minimum lap time	Crossings closer together than this are ignored.
Distance correction	Rescales each lap to the reference lap length.
Use GPS position inside the lap	Projects the position onto the reference trajectory when the fix is valid.
Manual Latitude / Longitude, Radius	Finish line for the *Manual Geographic Coordinates* source.
Cursor line color	Color of the vertical cursor line.
Update Server URL	Address of the update server (chap. 20).
Report logo	Image used in the reports.

18.2 Dataset colors and GPS coherence

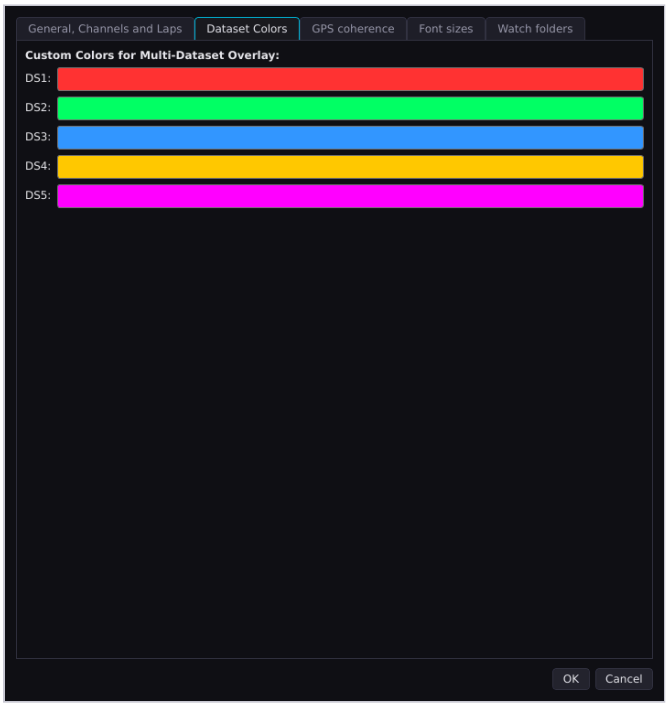


Figure 50 – Colors of datasets DS1–DS5.

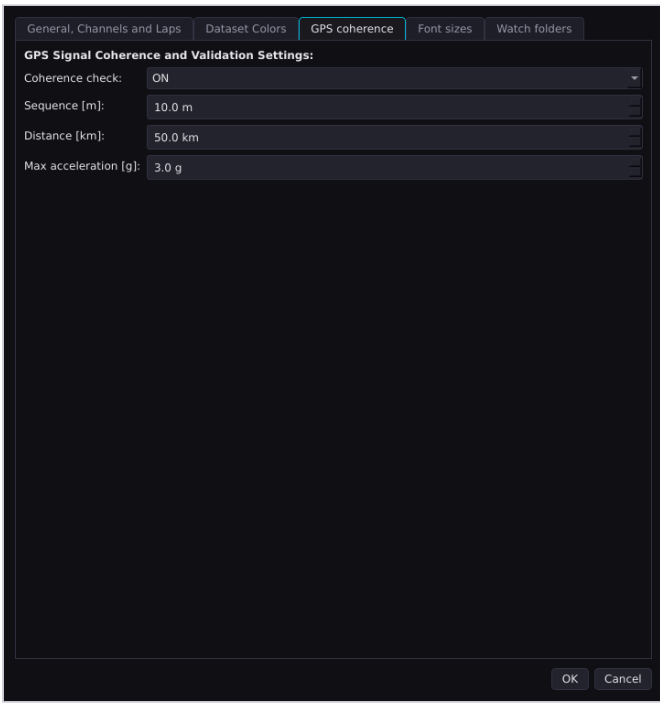


Figure 51 – GPS coherence parameters.

In the **GPS coherence** tab: **Coherence check** enables the check; **Sequence** is the minimum distance of coherent samples before the GPS is considered valid again; **Distance** is the maximum

deviation accepted from the expected position; **Max acceleration** is the acceleration beyond which a speed jump is considered an error.

18.3 Font sizes

The **Font sizes** tab sets the text size of the Channel Values Table, of the Lap Times table, of the AI assistant, of the Min-Max values and of the lap labels on the graph and of the numbers of the graph axes. **Default** uses the size of the interface; a smaller text leaves more room to the graphs. The sizes are applied when you press **OK**; **Reset to defaults** restores the initial values.

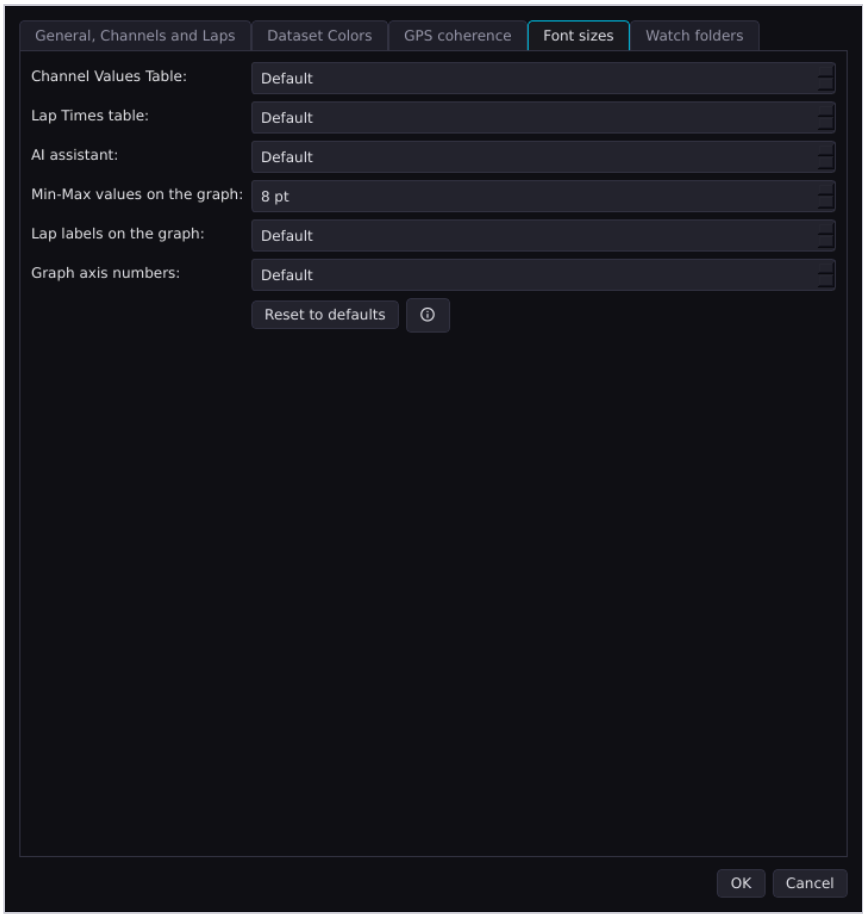


Figure 52 – The Font sizes tab.

19 Shortcuts and command palette

19.1 Default shortcuts

Keys	Action	Keys	Action
Ctrl + O / Ctrl + Shift + O	Open / add dataset	T / D	Time / distance base
Ctrl + W	Close selected dataset	+ - / Backspace	Zoom / undo zoom
Ctrl + I	Dataset information	F / L	Whole dataset / current lap
Ctrl + E / Ctrl + R	CSV export / report	← → (Shift ×10)	Cursor by one sample
Ctrl + P / Ctrl + Shift + S	Print / screenshot	Ctrl + ← →	Previous / next lap
Ctrl + L	Layout editor	Home End PgUp PgDown	Start, end, pan
Ctrl + S	Save channel layout	A / B / C	Marker A / B / clear
Ctrl + D	Lap comparison	S / Shift + S	Shift / reset shift
Ctrl + G	Add graph	1 … 5	Select dataset
Ctrl + M	Math channels	Y	AutoZoomY / ManualZoomY
Ctrl + J	AI Assistant	F5 … F10	Stats, Scatter, Histogram, FFT, 3D, Pie
Ctrl + K	Command palette	F1	Shortcut list
Ctrl + 1 … 9	Load layout #	F11	Full screen
Ctrl + T, Ctrl + Tab	New / next worksheet	Ctrl + Shift + L	Dark / light theme
Ctrl + Shift + A	ADV Compare	Ctrl + Shift + V	Add video
Ctrl + Shift + D	Dataset datasheet		

19.2 Customizing the shortcuts

Help → Keyboard Shortcuts (F1) lists all the commands by category with their shortcut and scope (*Whole window* or *Graph*). Double-click a command to change its shortcut; several keys are separated with `;`. The program reports conflicts. **Restore defaults** returns to the initial configuration.

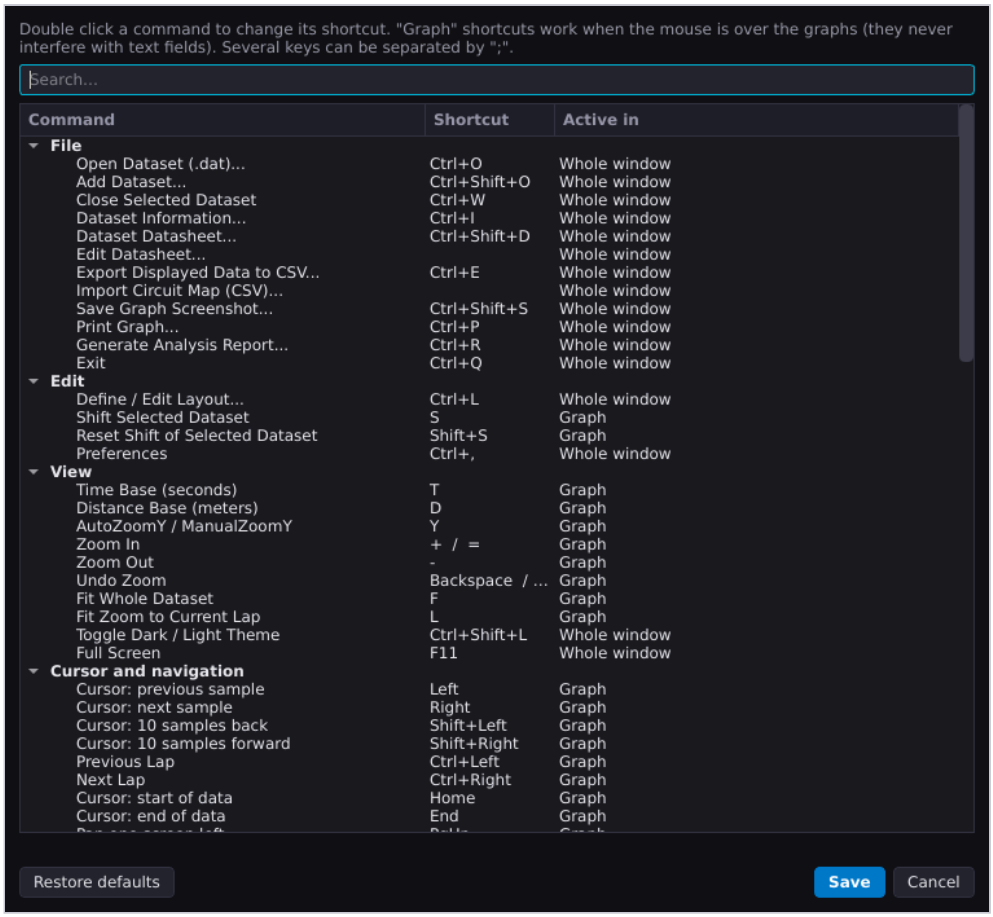


Figure 53 – The keyboard shortcuts window.

19.3 Command palette

`Ctrl`+`K` opens a search box covering all the program commands and the laps of the open datasets: type a few letters and press `Enter`. It is the fastest way to find a function without looking for it in the menus.

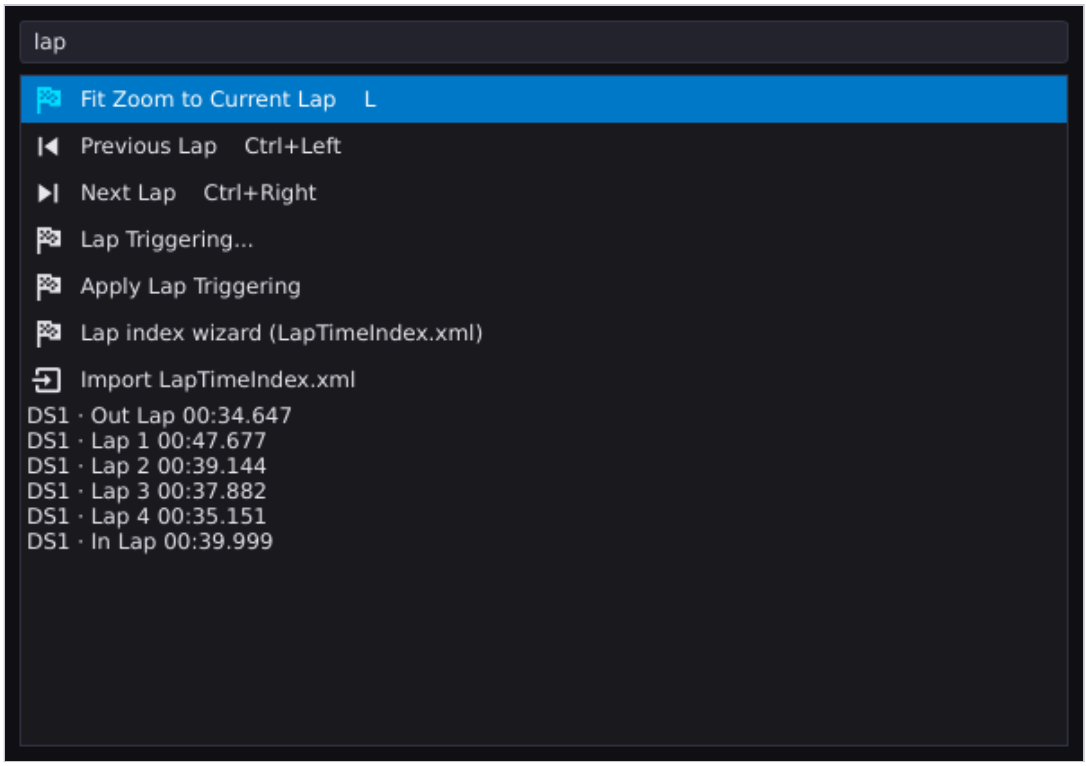


Figure 54 – The command palette: searching for "lap" shows the wizard, the index import and the laps.

20 Updates

Help → Check for Updates… checks whether a new revision is available on the Gubellini Electronics server (address in the Preferences).

1. The program reads the `versione_software.txt` file on the server and compares its revision number with the installed one (025).
2. If the revision on the server is newer, you are offered the download of `GUBELLINIDataAnalysis.zip`, which takes place in the background into the user's *Downloads* folder.
3. If `GUBELLINIDataAnalysis.zip.sha256` is present on the server, the integrity of the downloaded file is verified: a corrupted or tampered file is discarded.
4. Finally, install the new revision with the installer contained in the package. Layouts and preferences are preserved.

The installed revision is shown in **Help → About GUBELLINI DataAnalysis**.

21 Laps, sectors and track

Chapters 21–25 describe the functions introduced in Revision 025, developed in GUBELLINI RaceLab and brought to DataAnalysis. The channels of the GUBELLINI datasets keep their hardware input id (GPS3, CAN2, ANALOG1…): the automatic functions recognise speed, engine speed, throttle, brake, suspensions… from the **name** written in `Channels_details.xml`.

21.1 Sectors, ideal lap and rolling lap

The **Sectors** window (next to Lap Times, or from **Windows**) cuts the lap into sectors independent of the split times of the logger.



Figure 55 – The sector boundaries on the graph (dashed lines) and on the map.

GPS and metre sectors

- **GPS points**: every boundary is a point of the track (latitude, longitude). Every lap is measured on the same piece of track even on a wider line or with the finish line taken a little earlier or later: the most reliable comparison.
- **Metres from the finish line**: every boundary is a distance from the finish line; works without GPS too.

Set places the boundaries in the middle of the longest straights (away from the finish line) or at equal distances. Sectors are saved per circuit.

Moving the boundaries

With **On the graph** on, the boundaries appear as dashed lines for every lap of the selected dataset: drag one to move the boundary for all laps. The values can also be typed in the table of the boundaries.

Ideal and rolling lap

The table shows the sector times of every lap: the best sector in violet, the best lap in red. Below, the **ideal lap** (sum of the best sectors) and the **rolling lap** of every dataset: the fastest lap measured from any boundary, with the sectors of two consecutive laps.

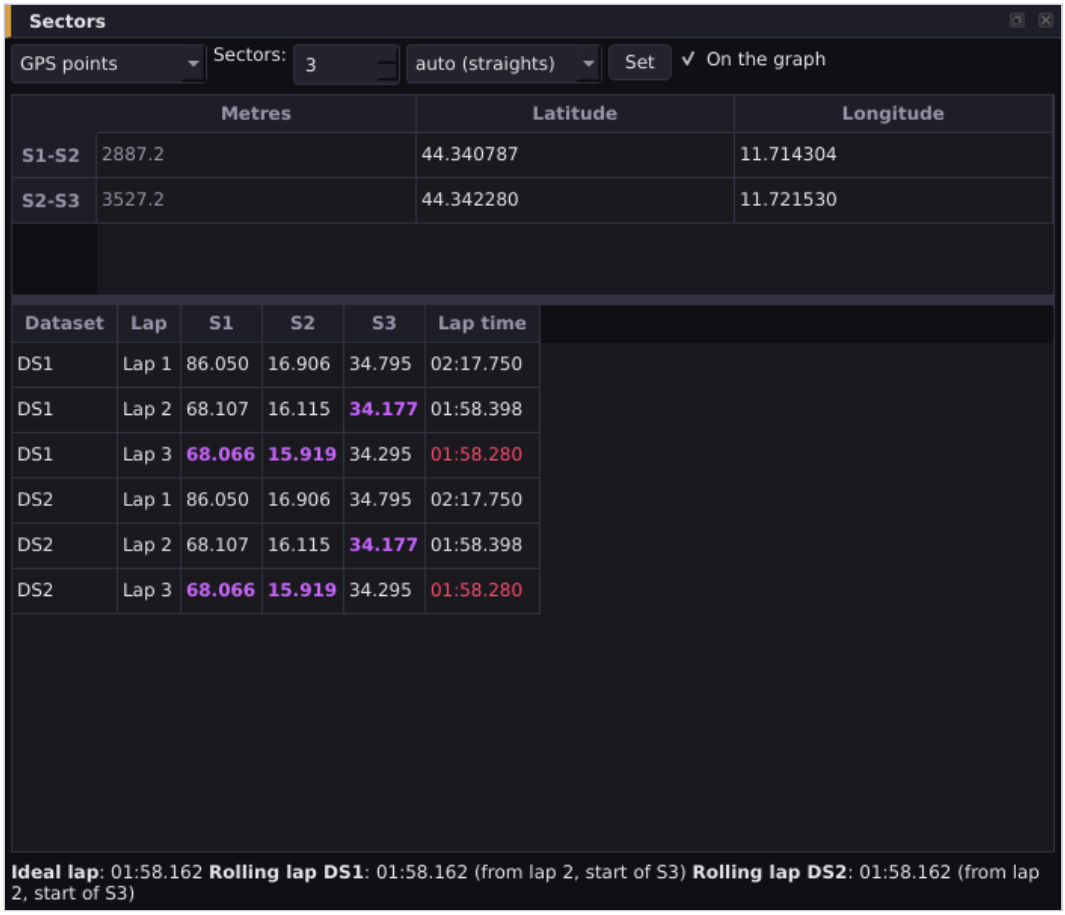


Figure 56 – The Sectors window.

21.2 Circuit map: strip, satellite, colour bands and advantage

The checkboxes and the ⚙ menu of the **Circuit Map** window add four tools.

Track strip

The track is drawn as a strip of its real width, so the GPS trajectory shows roughly where the rider goes. **Track width: Auto** chooses it from the lap length (6–8 m for kart tracks, 11–13 m for circuits); it can be set by hand. Without a CSV map the strip follows the GPS reference lap.

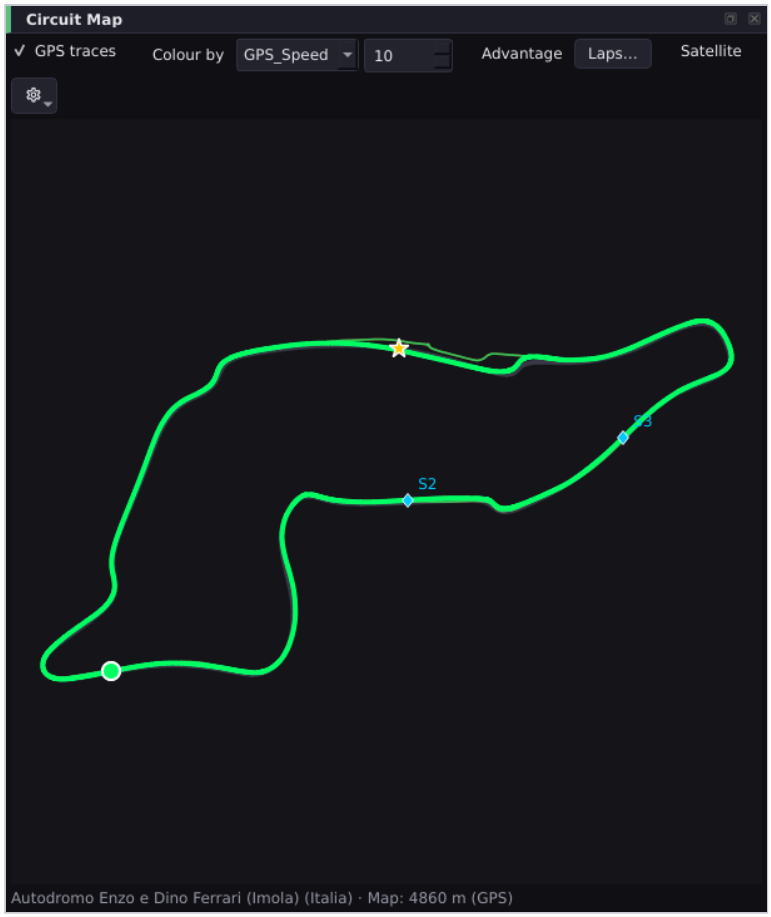


Figure 57 – The track strip and the GPS trajectory.

Satellite image

The **Satellite** box puts the satellite image of the circuit under the track, placed with the GPS coordinates. Internet is needed the first time; the images are kept on disk. The image service is set in **Preferences → Satellite tiles**.

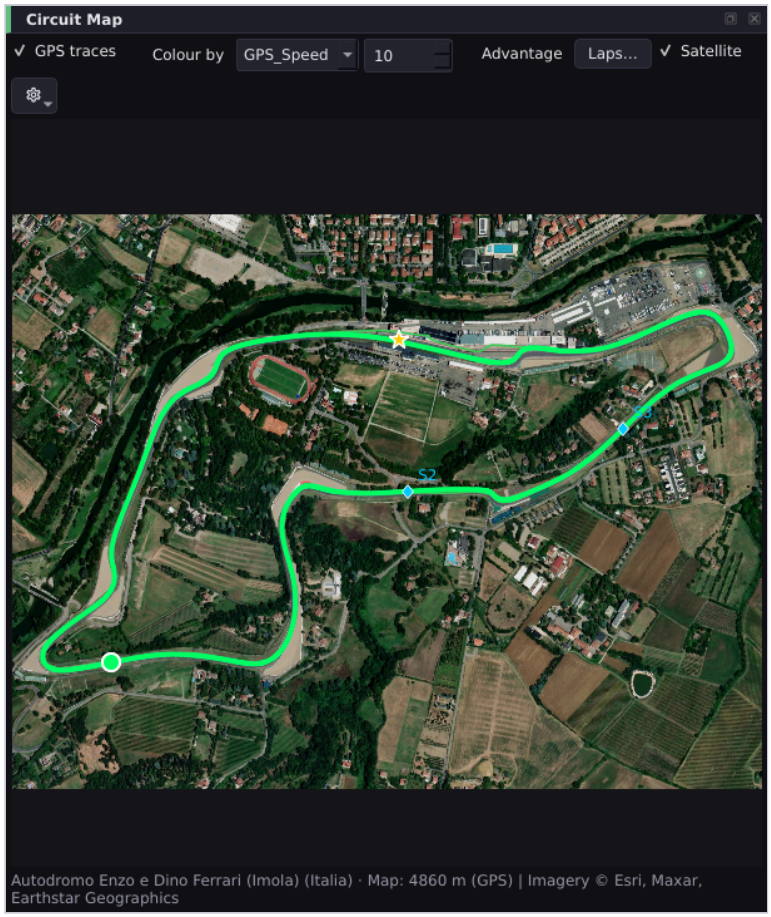


Figure 58 – Satellite image under the trajectory.

Colour bands

Colour by and a channel colour the trajectory of the selected dataset, in the visible window, by the value of the channel (10 bands by default, from blue for low values to red for high ones), with the legend under the map.

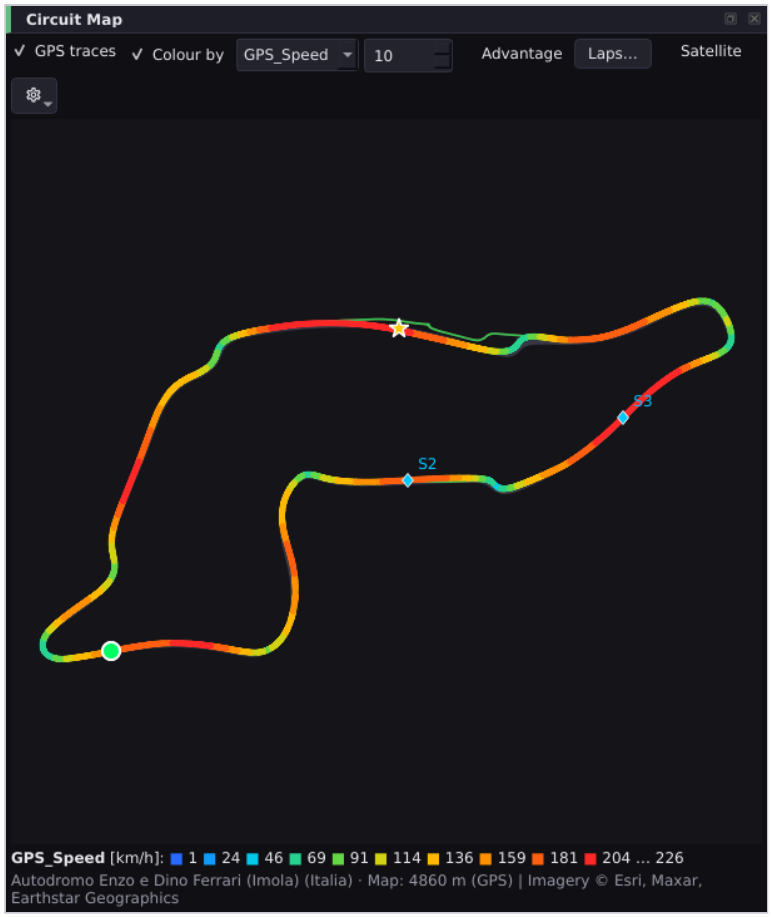


Figure 59 – Trajectory coloured by the speed.

Advantage

With **Advantage** every piece of the track gets the colour of the compared lap that covered it in the shortest time; under the map, the share of the lap in which each lap was ahead. By default the best lap of every dataset shown is compared (or the two best of a single dataset); choose the laps with **Laps…**.

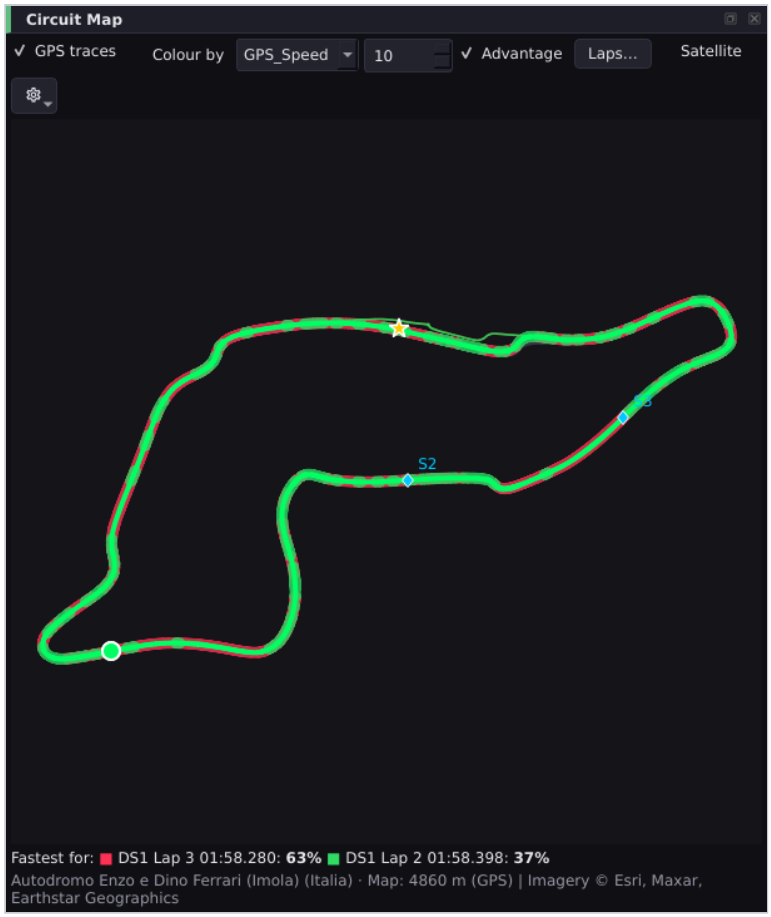


Figure 60 – The Advantage between the two best laps.

21.3 Common GPS finish line

Every logger finds its laps with its own beacon or finish line, so the laps of different files can start at different points of the track. **Tools → Common GPS Finish Line** applies one finish line to every open dataset and to the ones added later in the session. The finish line can be the position of the selected dataset at the cursor, the finish line of the circuit map or the one of a dataset; latitude, longitude and radius can be corrected. Datasets that never pass near the point (another circuit) are left as they are.

21.4 Circuit recognised also without laps

When the file has no finish line (no GPS trigger in the logger configuration) and therefore no laps, at opening the circuit is recognised from the **whole GPS trajectory** among the maps of the `TrackMap` folder; then the laps are found with the **finish line of the catalog** `track_maps.xml` for the recognised circuit. A notification tells how many laps were found.

21.5 Lap Triggering (files without laps or GPS)

For a file without lap times, GPS position or beacon, the first time it is opened the program offers the **Lap Triggering** (also from **Tools → Lap Triggering**):

1. The window shows the speed trace against the distance.
2. The yellow **Offset** line is the first finish line crossing; the violet lines follow every **lap distance**. Type the two values or set them by dragging the lines.
3. **Estimate from the speed** finds the distance after which the trace repeats itself.

4. Choose the circuit map: the vehicle is placed on it by its distance from the finish line.
5. Check that every violet line falls on the same point of the lap and press **Apply**.

The values stay with the file and are applied every time it is opened. For the next file with the same problem the program offers **Apply Lap Triggering** with the same values.



Figure 61 – Lap Triggering: Offset (yellow) and lap distances (violet).

The same question still offers, as before, the **guided procedure** that creates the LapTimeIndex.xml file and the import of a LapTimeIndex.xml (chapter 6).

21.6 Rider sheet

Tools → Rider Sheet prepares a vertical 1080×1920 image (a phone screen) or a PDF to send to the rider: best lap, ideal and rolling lap, sectors with the time to gain, map coloured where the lap is faster or slower than the reference, speed over the lap and **key points** (early braking, minimum corner speed, throttle opening). The reference is the best lap of another dataset (a team mate) or, with one dataset, the second best lap. The key points can be edited or written by the AI assistant; **Copy** puts the image on the clipboard to paste it in WhatsApp.

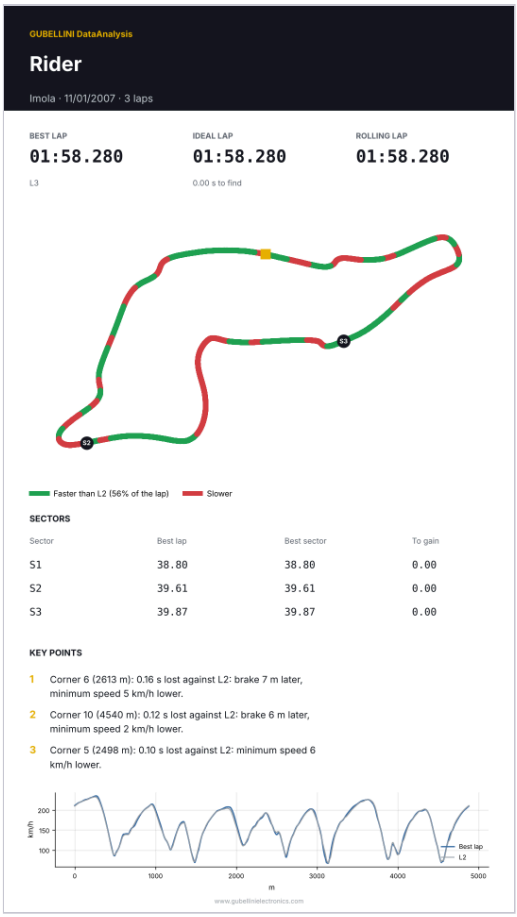


Figure 62 – The rider sheet (Yamaha R6 at Imola).

22 Engine and vehicle

22.1 Fuel map

Tools → Fuel Map puts the data on an RPM × TPS grid, like the map of the control unit: in every cell the mixture measured by the probe and the correction to reach the target. Every sample is shared among the four cells around it, as the ECU reads its map.

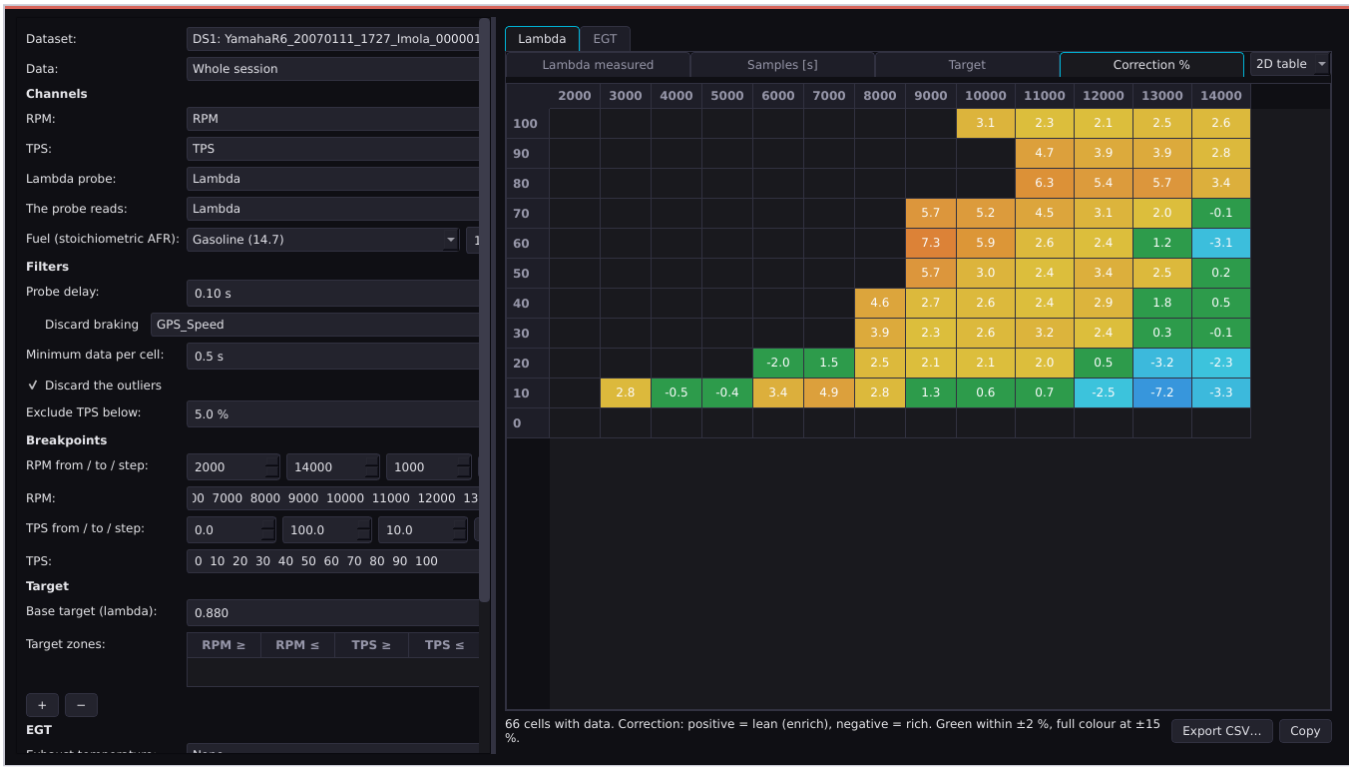


Figure 63 – The correction per cell: green in target, yellow/red lean, cyan/blue rich.

Settings

- **Channels**: RPM, TPS (or None), lambda probe; the probe can read lambda or AFR.
- **Fuel**: with its stoichiometric AFR (gasoline 14.7, E85 9.8, methanol 6.4…) or custom.
- **Probe delay**: the value read is matched with RPM and TPS of this time before.
- **Discard braking**, **Minimum data per cell**, **Discard the outliers**, **Exclude TPS below** (rows shown pale and not corrected).
- RPM and TPS **breakpoints** generated with from/to/step or typed by hand; base **target** and zones with a different target.
- **Data**: whole session, lap at the cursor or visible window.

Readings

The sheets show the measured mixture, samples (seconds per cell), target and **Correction %**: positive when the mixture is lean and must be enriched, negative when rich. Green within ±2 %, full colour at ±15 %. The views are 2D table, 3D surface and curve per TPS row. The **EGT** tab shows the highest exhaust temperature per cell and the difference from the minimum and maximum target.

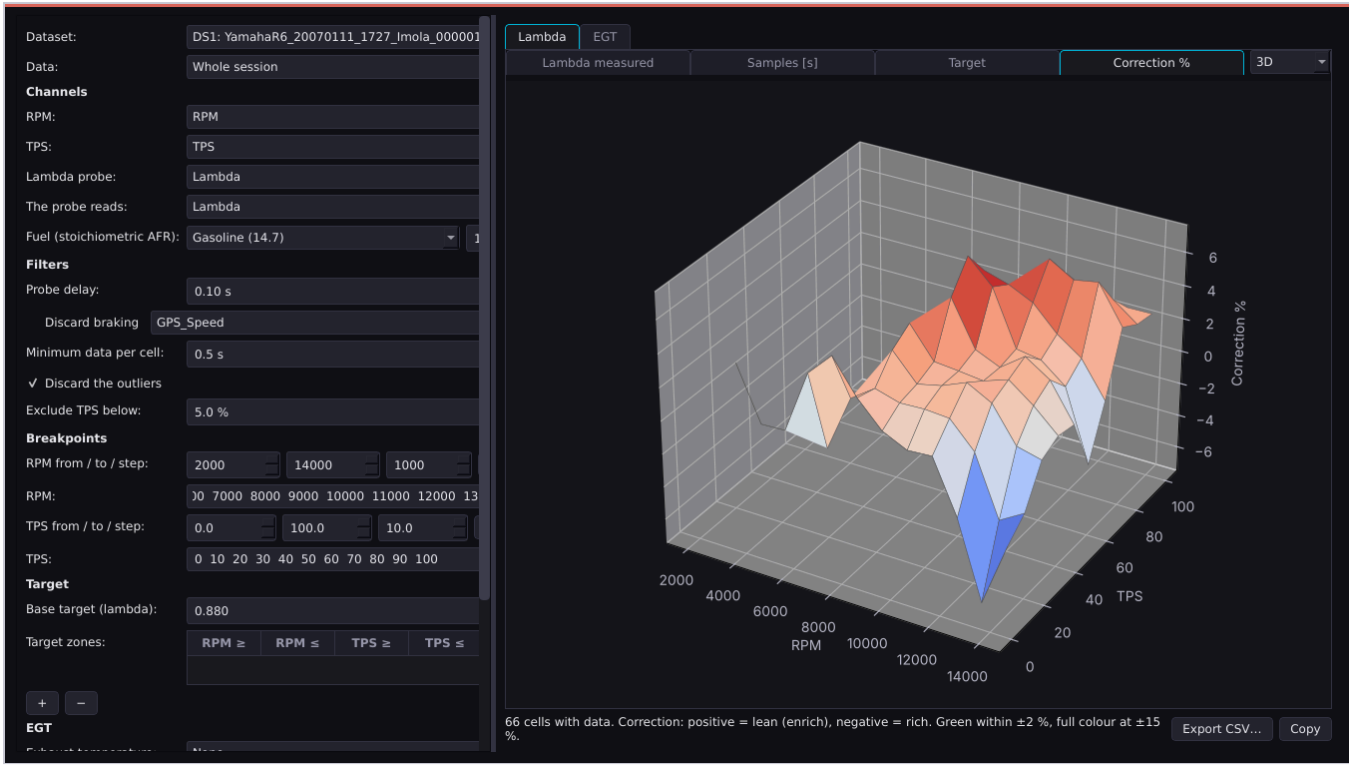


Figure 64 – The correction as a 3D surface.

ECU map

Import map… reads the ECU map from a CSV (fuel / injection times, which is multiplied, or corrections in percent, which are added): its rows and columns become the breakpoints and the Original map and Corrected map sheets appear. **Export corrected map…** writes the file in the same layout, corrected only in the cells with data.

22.2 Health check

Every time a dataset is opened the program checks the secondary channels nobody looks at in the rush between two sessions: **battery voltage** (level and drops), the **limits** of the thresholds (temperatures, oil pressure under load…), **clutch slip** (engine speed rising more than the gear engaged), the **sensors** (no signal while the vehicle moves, missing samples, isolated spikes) and the **GPS** quality. When something needs a look a notification appears. **Tools → Health Check** (`Ctrl`+`H`) lists the findings of every dataset, the most serious first, with channel, lap and time; a double click takes the graphs there and shows the channel. A finding is a reason to look at the channel, not a diagnosis.



Figure 65 – The Health Check window.

22.3 Thresholds

Tools → Thresholds holds the limits used by the health check, the suspension analysis and the AI assistant, with a tab per group (engine, electrical, transmission, suspension, brakes, chassis and GPS, other). Every row has a channel – a quantity recognised from the channel names (water temperature, oil pressure…) or the name of a channel – with warning (min / max), critical (critical min / max) and when it applies (always, vehicle running, engine under load); an empty value is not checked. The **analysis parameters** (damper speed boundary, clutch tolerance, battery drop…) are at the bottom. Several sets can be kept, one per vehicle: the active one is used. The thresholds are saved in `Profiles/thresholds.json`.

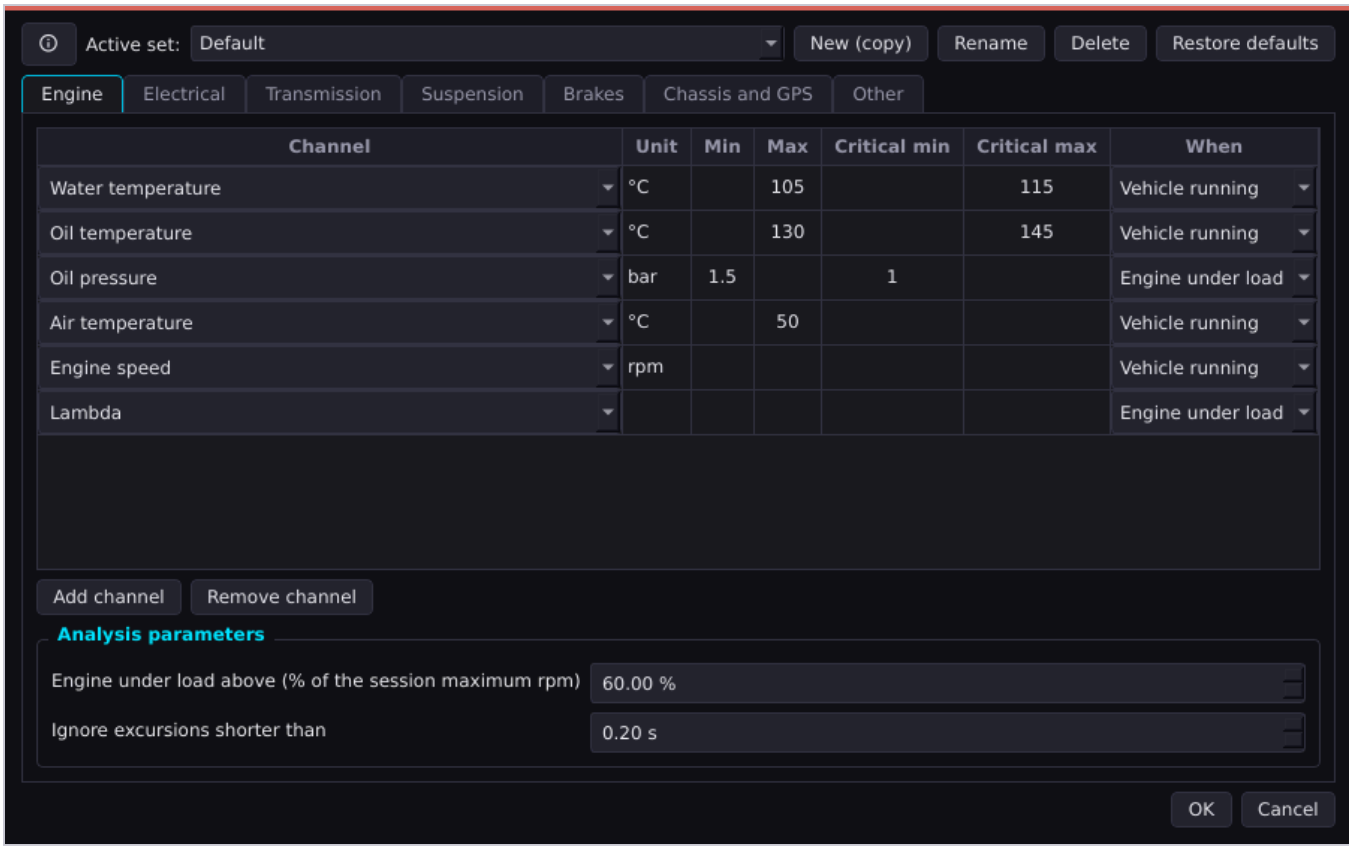


Figure 66 – The thresholds of the Engine group.

22.4 Suspension analysis and corner by corner comparison

From the suspension position channels (recognised by name: fork, shock, susp…) the **suspension analysis** computes the damper speed and its histogram: time in compression and rebound at low and high speed, symmetry, end of travel. The **corner by corner comparison** splits two laps into corners and for each one compares braking point, minimum speed, point where the throttle is fully open again and time lost. They are used by the AI assistant (**Suspension** and **Engineer brief** buttons) and by the rider sheet.

23 AI assistant: commands and context

The AI assistant (chapter 16, with the API key of your provider) can now act on the program and knows the context of the project.

23.1 Commands on the program

While answering, the assistant can **show channels** in the main graph, **add a graph** or a **panel** (FFT, scatter, g-g, alarms, values, gauge), **highlight a stretch** it talks about, **change the selected dataset** and **create a worksheet**. It computes health check, suspension analysis and corner comparison with the functions of the program, not by eye. The commands on the interface can be turned off in the assistant setup.

23.2 Context and sensors

Tools → AI Context and Sensors: how the data is used (vehicle, championship, what you are looking for) and a description for the channels the name does not explain (where the sensor is mounted, sign conventions, meaning of a status value). The text is sent with every question and stays in the settings of this computer. The *Recognised as* column shows what the program understands from the name by itself.

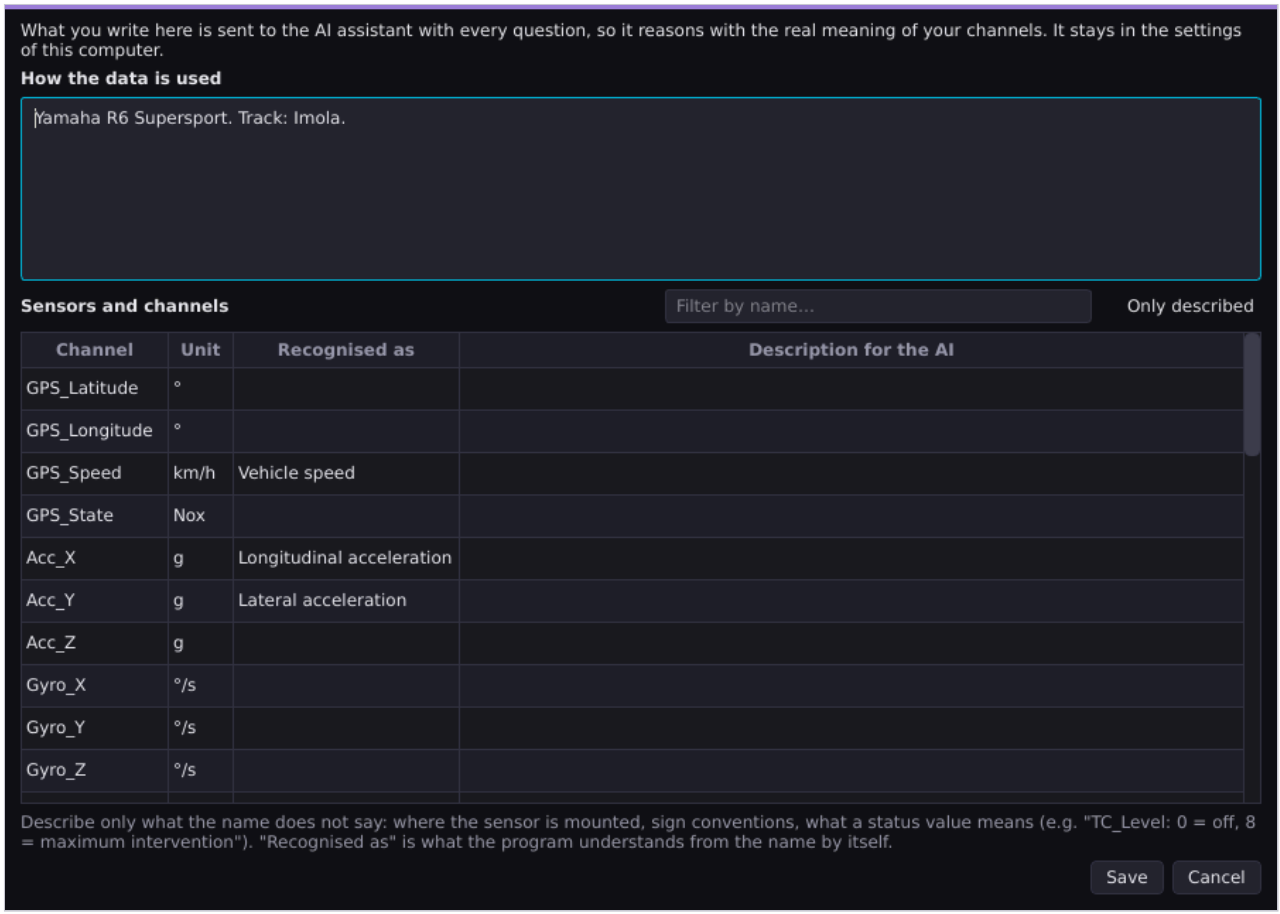


Figure 67 – Context and sensors for the AI assistant.

23.3 Quick buttons and frequent requests

Above the question box, three **quick buttons**: **Engineer brief** (where time is lost, corner by corner), **Health check** (of the open datasets) and **Suspension**. Below, **my frequent requests**: four buttons written once with ✎ (button name and request). The thresholds (chapter 22.3) are always in the instructions of the model.

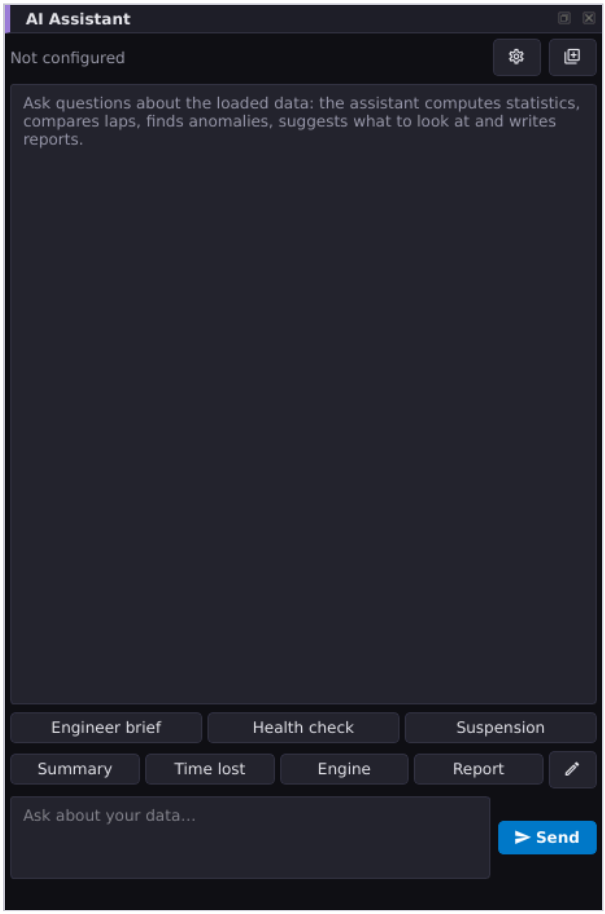


Figure 68 – Quick buttons and frequent requests.

24 Worksheets and interface

24.1 Worksheet panels

Below the graphs every worksheet can hold up to eight panels (**+ Panel**): FFT, Scatter, G-G diagram, Alarms, Channel values, Gauge. They follow cursor and zoom of the graphs and are saved with the worksheet.



Figure 69 – Channel values, Scatter and Gauge panels with two datasets.

Automatic and manual axes

FFT, Scatter and G-G have the **A** button: on, the axes follow the data; off, the limits written in the panel settings (X/Y minimum and maximum) hold, or the view as it is now when none is written.

More channels and more datasets

- **Scatter**: one X channel and one or more Y channels.
- **G-G**: up to three pairs of accelerations (e.g. those of the inertial platform and those computed from the GPS).
- **Channel values**: several channels, one column per dataset.
- With more datasets shown every panel shows all of them in their colours (the Gauge with one needle per dataset); with one dataset the channels have their own colours.

24.2 Opening, adding and hiding datasets

- **Open Dataset** replaces the selected dataset (which keeps its DS number); with no dataset open it becomes DS1.
- **Add** loads one more dataset, up to five.

- With more datasets, a **tick** in front of every dataset in the status bar shows or hides it for the moment in graphs, maps and panels. With one dataset shown the curves get back the colours of the channels.



Figure 70 – DS2 hidden with its tick: the graph is back to the channel colours.

24.3 Faster cursor and zoom

The graphs draw one minimum and one maximum point per pixel of the visible part, the traces of the maps are kept in memory and the windows follow the cursor at most once per frame: moving the cursor and zooming stay immediate also with several long datasets. In **Preferences → General**, **Graph detail** chooses *Screen resolution* (no visible difference) or *Fast* (half the points, for slow computers); the data and the values do not change.

24.4 GPS trajectory map with every dataset

The **GPS Trajectory** window shows the trajectory of every visible dataset in its colour, each with its point at the cursor (taking the shift of the datasets into account).

24.5 A more compact interface

- The **left column** (Channels, Laps, Sectors) can be made narrow: the commands wrap and the columns of the Channels table fit the widest text they can show.
- **Less text on screen**: short window titles (Channels, Laps, Circuit Map, GPS Trajectory, Alarms), text in the toolbar only on the main commands (the name of the others is in the tooltip), the secondary commands of the circuit map in the ⚙ menu and the explanations behind the ⓘ button.
- **Colours by working area**: a coloured strip on the title of the docked windows and a top edge on the windows of the area – values (light blue), laps (amber), track (green), vehicle (red), AI

(purple) – in the dark and light theme. The icons follow the theme change.

- **New names of the windows and buttons**: Channel Values Table → Channels; Lap Times and Outings → Laps; Virtual LEDs / Alarms → Alarms; Circuit Map (CSV Normalized) → Circuit Map; GPS Trajectory Map → GPS Trajectory; Time Base [s] → Time [s]; Distance Base [m] → Distance [m].

25 Data and workflow

25.1 Editing the datasheet

The datasheet stays the one associated with the dataset by GUBELLINI DataStudio at the download (`Datasheet.xml` in the dataset folder, section 4.5). **File → Edit Datasheet** changes its **values** (driver, tyres, setup…): the file keeps the DataStudio format, with dataset information, template and **part counters** unchanged. **Copy previous values** takes the values from the datasheet of another dataset of the same folder (not the dates).

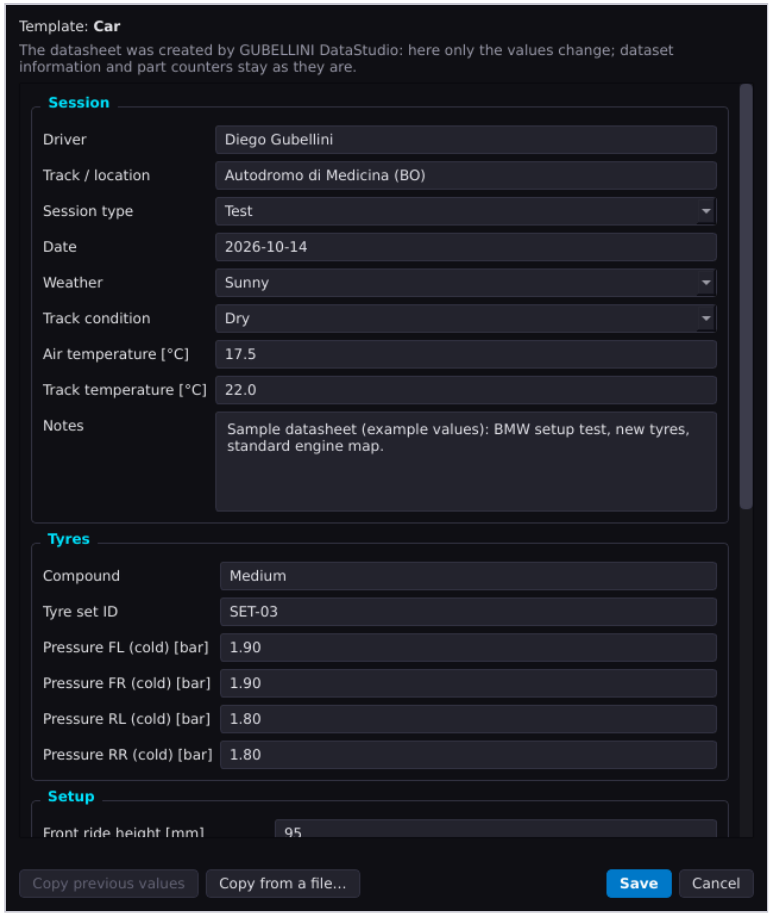


Figure 71 – Editing the datasheet of the Medicina dataset.

25.2 Report a problem

Help → Report a Problem prepares a message for GUBELLINI support (`dataanalysis@gubellinielectronics.com`, subject `[DataAnalysis] …`): kind (bug, file not opened correctly, suggestion), description, steps to reproduce and, if you want, the diagnostics (revision, system, format and length of the open datasets). It holds no data and no file paths and has no attachments. **Send by email** opens your email program with the message ready: check it and send it from there.

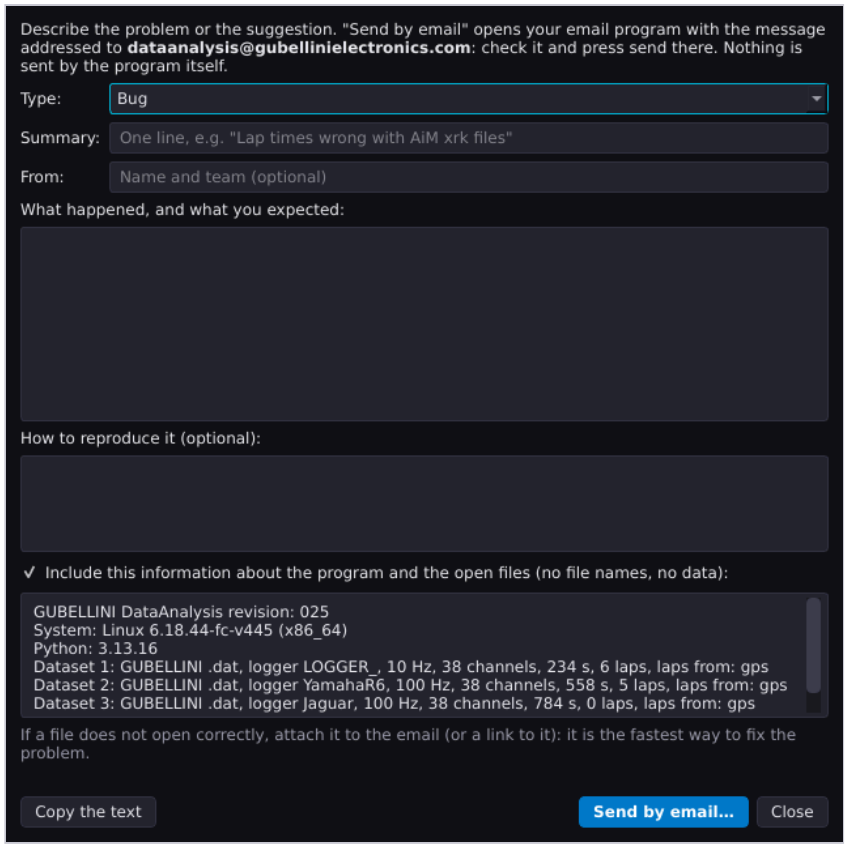


Figure 72 – Report a problem.

25.3 Watched folders

In **Preferences → Watch folders** list the folders where the datasets arrive (e.g. the one where GUBELLINI DataStudio downloads the data from the logger). With **Watch the folders** on, the program checks them every few seconds and, when it finds a new complete dataset, shows a notification with the **Open** and **Add** buttons. Datasets that arrived while the program was closed are reported at the next start.

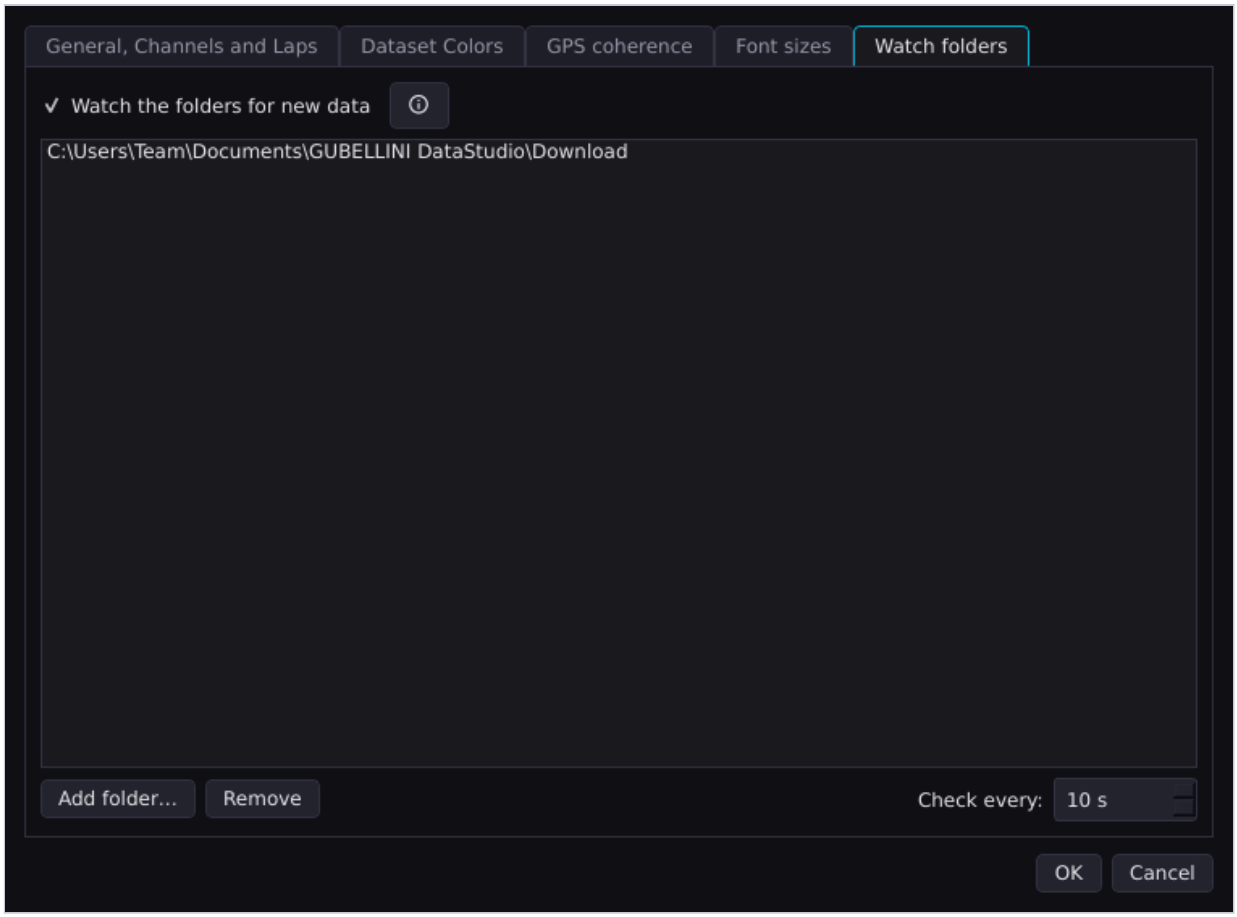


Figure 73 – The watched folders in the Preferences.

26 Troubleshooting

Problem	Cause and solution
No laps detected	The GPS has no valid fix or the finish line coordinates are wrong. Check the status bar (*GPS ✗*) and the GPS Trajectory Map; use a beacon channel, import LapTimeIndex.xml or use the wizard (§6.4).
Laps with strange or double times	Increase the **Minimum lap time**; for the manual finish line, reduce the detection radius.
The position on the map does not match	Map not aligned: use **Map from GPS** to create a georeferenced map, or adjust **Finish offset**.
Distance differs from one lap to another	Enable **Distance correction** and set the **Fallback speed channel** for GPS dropouts.
A channel looks flat	In ManualZoomY the layout limits are too wide: press Y for AutoZoomY or correct Min/Max in the layout.
Single-key shortcuts do not work	The text cursor is in an input field: move the mouse over the graph.
Channel names such as GPS3, CAN2	`Channels_details.xml` is missing from the dataset folder: copy the folder generated by the logger next to the .dat file.
Windows have disappeared or are disordered	**Layouts → Restore Default Windows Layout**; closed windows can be reopened from the **Windows** menu.
The AI assistant does not answer	See §16.4.
Check for updates fails	Check the connection and the server URL in the Preferences.

Technical support
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